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# FedKLS: KL-Guided Spectral Allocation for Large-Model Federated Learning in Heterogeneous IoT Environments

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# FedKLS: KL-Guided Spectral Allocation for Large-Model Federated Learning in Heterogeneous IoT Environments

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**Abstract**—Large pre-trained models are increasingly adopted in intelligent Internet-of-Things (IoT) applications, but their deployment in federated environments is challenging due to high communication and computational costs associated with large-scale model adaptation. Moreover, severe non-IID data distributions across clients further complicate collaborative training, leading to degraded performance and limited personalization. To address these challenges, we propose FedKLS, a KL-guided spectral allocation framework for personalized federated learning. It leverages KL-divergence to quantify client heterogeneity and assigns each client to different regions of the singular value spectrum of a pretrained model for efficient low-rank adaptation. This allocation allows near-IID clients to exploit principal components, while highly non-IID clients adapt minor components of the spectrum, thereby improving robustness while preserving personalization for heterogeneous IoT devices. Extensive experiments across text, image, and multimodal benchmarks with ResNet, DistilBERT, Qwen, and CLIP ViT-B/32 backbones demonstrate that FedKLS consistently outperforms state-of-the-art parameter-efficient fine-tuning approaches, including LoRA, PiSSA, MiLoRA, full fine-tuning, and personalized federated methods such as pFedEdit and FedAS. Under highly non-IID settings ( $\alpha = 0.01$ ), FedKLS improves F1-score by up to 11–12% and reduces communication cost by approximately  $3\times$  compared to PiSSA. These results highlight the effectiveness of KL-guided spectral allocation for scalable and personalized federated adaptation of large-models. Our implementation is publicly available at: <https://github.com/phamkhanhquan197/FedKLS>.

**Index Terms**—Federated learning, IoT, large models, parameter-efficient fine-tuning, low-rank adaptation, personalization, non-IID data, dynamic data.

## I. INTRODUCTION

Federated learning (FL) has emerged as a promising paradigm for training models collaboratively across decentralized IoT devices while preserving data privacy [1], [2]. In FL, each client performs local training on its private data and only shares model updates with a central server for aggregation, without exposing raw data.

However, extending FL to large pre-trained models introduces significant challenges due to the high communication and computation costs associated with large-scale model

adaptation. These costs become prohibitive in bandwidth-limited and energy-constrained IoT environments, where frequent transmission of large model updates is impractical. To address this challenge, parameter-efficient fine-tuning (PEFT) has emerged as a practical solution for adapting large models in FL. Instead of updating all model parameters, PEFT methods optimize only a small subset while freezing the backbone, significantly reducing communication and storage overhead. Among these approaches, Low-Rank Adaptation (LoRA) [3] introduces trainable low-rank matrices into model layers, enabling efficient updates with minimal parameter exchange. Subsequent works, such as QLoRA [4] and LoRA-FA [5], further improve efficiency by incorporating quantization or reducing activation-related overhead during fine-tuning, making them particularly suitable for resource-constrained and communication-limited environments. These advances establish PEFT as a scalable solution for large-model adaptation in FL.

Nevertheless, existing PEFT methods are primarily designed for centralized settings and implicitly assume a shared adaptation structure across all clients, without explicitly accounting for heterogeneous data distributions in federated environments. In federated settings, FL systems face complex data heterogeneity across edge devices, involving both spatial variance—such as non-IID data across sensing modalities [6]—and temporal dynamics, as local data evolve over time due to concept drift or changing environmental contexts. This combined spatial and temporal heterogeneity severely degrades performance and slows down convergence. Prior work, such as FedProx [7], SCAFFOLD [8], and MOON [9], has shown that heterogeneity across clients can result in client drift and unstable optimization. While personalized federated learning has been extensively studied, including methods such as pFedEdit [10], FedAS [11], pFedMe [12], FedPer [13], and Ditto [14], these approaches primarily focus on parameter-level personalization and do not explicitly leverage the structural properties of large models for heterogeneity-aware adaptation.

To address these limitations, we propose **FedKLS**, a KL-guided spectral allocation framework for personalized federated learning tailored for large-model-driven IoT systems. FedKLS leverages the Kullback–Leibler (KL) divergence to quantify the discrepancy between each client’s local data and a reference IID distribution. By periodically updating these KL divergence values across communication rounds, FedKLS captures both static statistical heterogeneity and temporal distribution shifts. This heterogeneity signal is then mapped to different regions of the singular value spectrum to guide spec-

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tral allocation. Specifically, clients with near-IID data exploit principal components to promote global generalization, while highly skewed or evolving clients focus on minor components to enhance personalized performance.

Through this mechanism, FedKLS realizes structural personalization by allocating distinct spectral subspaces to different clients, enabling each client to align model updates with its local data distribution while still benefiting from shared global knowledge. This continual, heterogeneity-aware spectral allocation ensures stable convergence and robust performance across diverse and dynamically changing IoT environments.

Extensive experiments on text, image, and multimodal benchmarks using ResNet [15], DistilBERT [16], Qwen [17], and CLIP ViT-B/32 [18] backbones demonstrate that FedKLS consistently outperforms state-of-the-art PEFT methods, including LoRA, PiSSA, MiLoRA, full fine-tuning (FFT), and personalized federated approaches such as pFedEdit [10] and FedAS [11]. Under dynamic data scenarios that emulate real-time IoT streams, FedKLS maintains stable convergence and strong performance, highlighting its practicality for real-world deployments.

Our main contributions are summarized as follows:

- We analyze the limitations of existing PEFT methods in federated IoT settings and reveal how spectral components correspond to generalizable and client-specific representations under heterogeneous data distributions.
- We propose FedKLS, a KL-guided spectral allocation framework for personalized federated learning that explicitly links client data heterogeneity to spectral subspace selection via KL-divergence, enabling dynamic and structured personalization for large models.
- We demonstrate that FedKLS significantly improves communication efficiency and convergence stability, outperforming state-of-the-art PEFT and personalized FL baselines in heterogeneous IoT environments.

## II. BACKGROUND AND RELATED WORK

### A. Overview of Singular Value Decomposition (SVD) and Low-Rank Matrix Approximations in Model Adaptation

Singular Value Decomposition (SVD) is a fundamental matrix factorization technique that is widely used to decompose and compress neural network weight matrices. It decomposes a matrix  $W \in \mathbb{R}^{m \times n}$  into three components:  $U \in \mathbb{R}^{m \times r}$ , a matrix of left singular vectors;  $\Sigma \in \mathbb{R}^{r \times r}$ , a diagonal matrix of singular values; and  $V^\top \in \mathbb{R}^{r \times n}$ , a matrix of right singular vectors, where  $r \ll \min(m, n)$ . This decomposition,

$$W = U\Sigma V^\top, \quad (1)$$

where the singular values in  $\Sigma$  are arranged in descending order.

In recent years, the rise of LLMs with billions of parameters has presented new challenges for efficient model adaptation and fine-tuning. Traditional full-parameter updates become computationally expensive and often impractical, motivating the development of PEFT methods. One of the most widely

used PEFT methods is LoRA. Importantly, the connection between SVD and LoRA can be made explicit: if we set

$$A = U_r \sqrt{\Sigma_r}, \quad B = \sqrt{\Sigma_r} V_r^\top, \quad (2)$$

then

$$AB^\top = U_r \Sigma_r V_r^\top, \quad (3)$$

which is exactly the truncated SVD low-rank approximation. In practice, LoRA generalizes this idea by introducing  $A \in \mathbb{R}^{m \times r}$  and  $B \in \mathbb{R}^{n \times r}$  as learnable low-rank matrices, instead of fixing them to singular vectors. The low-rank update is expressed as

$$W' = W + \frac{\alpha}{r} \Delta W = W + \frac{\alpha}{r} AB^\top. \quad (4)$$

where  $\alpha$  is a scaling factor that controls the contribution of the low-rank update. This reparameterization dramatically reduces the size of the trainable parameters from  $\mathcal{O}(m \times n)$  to  $\mathcal{O}(r(m+n))$ , while preserving the representational feature needed for downstream tasks.

### B. Spectral-Based Low-Rank Adaptation Methods

Building on LoRA's SVD-inspired foundation established above, many recent works have investigated how different spectral components can improve adaptation. For example, DoRA [19] decomposes weights into magnitude and direction, VeRA [20] employs fixed random matrices for efficient adaptation, AdaLoRA [21] dynamically allocates rank during training, and QLoRA [4] applies quantization to reduce memory consumption. Although these methods are effective in centralized settings, they are not designed to address federated learning challenges such as client heterogeneity or personalization.

Among spectral-based approaches, LoRA, PiSSA, and MiLoRA are particularly relevant to our study because they explicitly leverage the SVD structure of pretrained weights, which can be extended to federated environments. Their theoretical motivation is supported by SoMA [22], which shows that principal singular components tend to encode generalizable features, whereas minor components capture domain-specific information.

Fig. 1 illustrates the adapter initialization strategies of PiSSA, MiLoRA, and the proposed FedKLS based on the SVD of the pre-trained weight matrix  $W = U\Sigma V^\top$ . In Fig. 1a, **PiSSA** initializes the low-rank adapters using the principal singular subspace. It extracts the dominant components  $(U_p, \Sigma_p, V_p)$  and constructs the adapters as

$$A_p = U_p \sqrt{\Sigma_p}, \quad B_p = \sqrt{\Sigma_p} V_p^\top. \quad (5)$$

In contrast, **MiLoRA** focuses on the minor singular subspace, as illustrated in Fig. 1b. It selects the smallest singular components  $(U_m, \Sigma_m, V_m)$  and constructs the adapters as

$$A_m = U_m \sqrt{\Sigma_m}, \quad B_m = \sqrt{\Sigma_m} V_m^\top. \quad (6)$$

These minor components capture complementary and task-specific information, which can enhance generalization under distribution shifts. While these spectral initialization strategies demonstrate the importance of different singular subspaces in model adaptation, both PiSSA and MiLoRA rely on a fixed spectral region shared across all data.

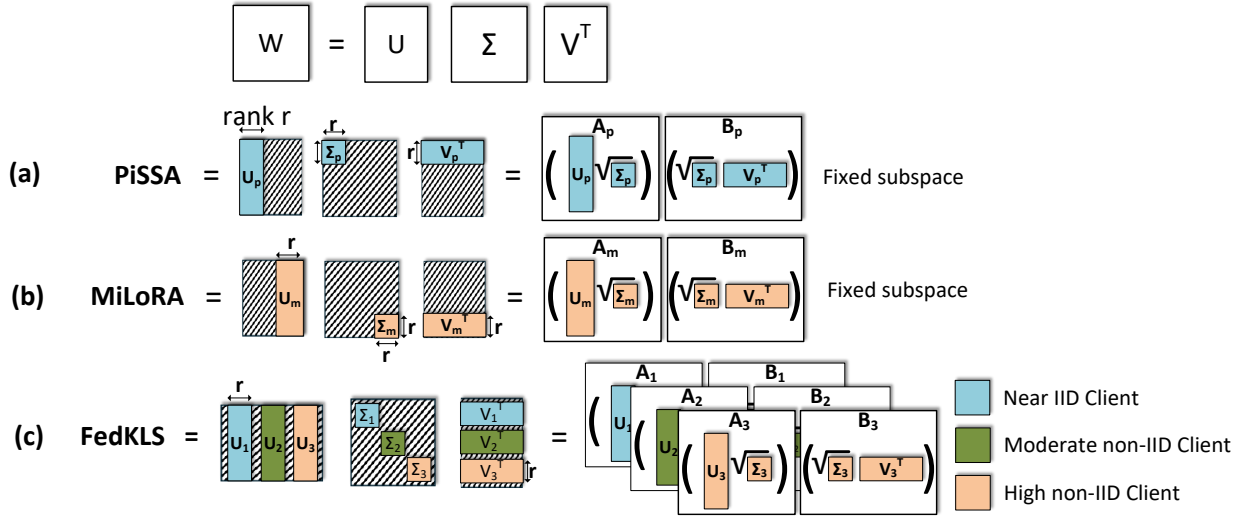


Fig. 1: Comparison among PiSSA, MiLoRA, and FedKLS. LoRA relies on random subspaces, PiSSA and MiLoRA utilize principal and minor subspaces, respectively, whereas FedKLS dynamically allocates client-specific subspaces based on the KL divergence.

### III. METHODOLOGY

#### A. FedKLS motivation

**Key Question:** *Why principal components are more suitable for near-IID clients and minor components for highly non-IID clients?*

Although spectral PEFT methods effectively exploit structured subspaces of pretrained weights, their spectral allocation typically assumes that a single spectral region is suitable for all training data. This assumption is often reasonable in centralized learning, where data are jointly optimized under a single distribution.

In FL, however, each client may follow a different local data distribution, resulting in heterogeneous update behaviors across clients. Such differences may lead to variations in the directions of local model updates in the parameter space.

As discussed in Section II-A, the SVD decomposition of the pretrained weight matrix  $W = U\Sigma V^T$  provides a spectral hierarchy over adaptation directions. Since the singular values in  $\Sigma$  are arranged in descending order, this ordering reflects the relative importance of different representation directions in the pretrained model. Principal components correspond to the singular vectors associated with the largest singular values in  $\Sigma$ , capturing the dominant structural patterns of the pretrained weight matrix, whereas lower singular components capture client-specific variations that are not aligned with dominant structures.

Under heterogeneous client distributions, enforcing a single fixed spectral slice across all clients may introduce structural mismatch. Clients with near-IID data typically share similar statistical characteristics with the global distribution, meaning that their local updates tend to align with the dominant directions of the pretrained weight matrix. These dominant directions correspond to the principal singular components of the pretrained model. Therefore, updates from near-IID clients

can be effectively represented within the principal spectral subspace.

In contrast, clients with highly skewed or non-IID data may generate updates that differ from these dominant directions. These directions are typically associated with smaller singular values, which correspond to minor spectral components capturing residual or task-specific variations. In such cases, adapting through minor spectral components can capture complementary or client-specific variations without significantly disturbing the dominant global representation.

Motivated by these observations, we propose **FedKLS**, which dynamically allocates spectral subspaces according to the degree of client heterogeneity. Instead of enforcing a single slice for all clients, FedKLS assigns different spectral regions according to the degree of client heterogeneity, enabling more flexible adaptation across heterogeneous data distributions.

Specifically, as illustrated in Fig. 1c, the SVD components are partitioned into multiple spectral intervals, such as  $(U_1, \Sigma_1, V_1)$ ,  $(U_2, \Sigma_2, V_2)$ , and  $(U_3, \Sigma_3, V_3)$ , corresponding to different levels of data heterogeneity. Clients with near-IID data primarily utilize the principal spectral segment, moderately non-IID clients adapt intermediate components, while highly non-IID clients focus on the minor spectral region. The corresponding low-rank adapters are constructed as

$$A_i = U_i \sqrt{\Sigma_i}, \quad B_i = \sqrt{\Sigma_i} V_i^T. \quad (7)$$

This heterogeneity-aware spectral selection enables FedKLS to dynamically balance global knowledge sharing and client-specific personalization.

#### B. FedKLS Architecture

Fig. 2 illustrates the overall architecture of FedKLS, which enables structural personalization through KL-guided spectral allocation. The framework first quantifies client heterogeneity using KL divergence computed locally at each client. At the

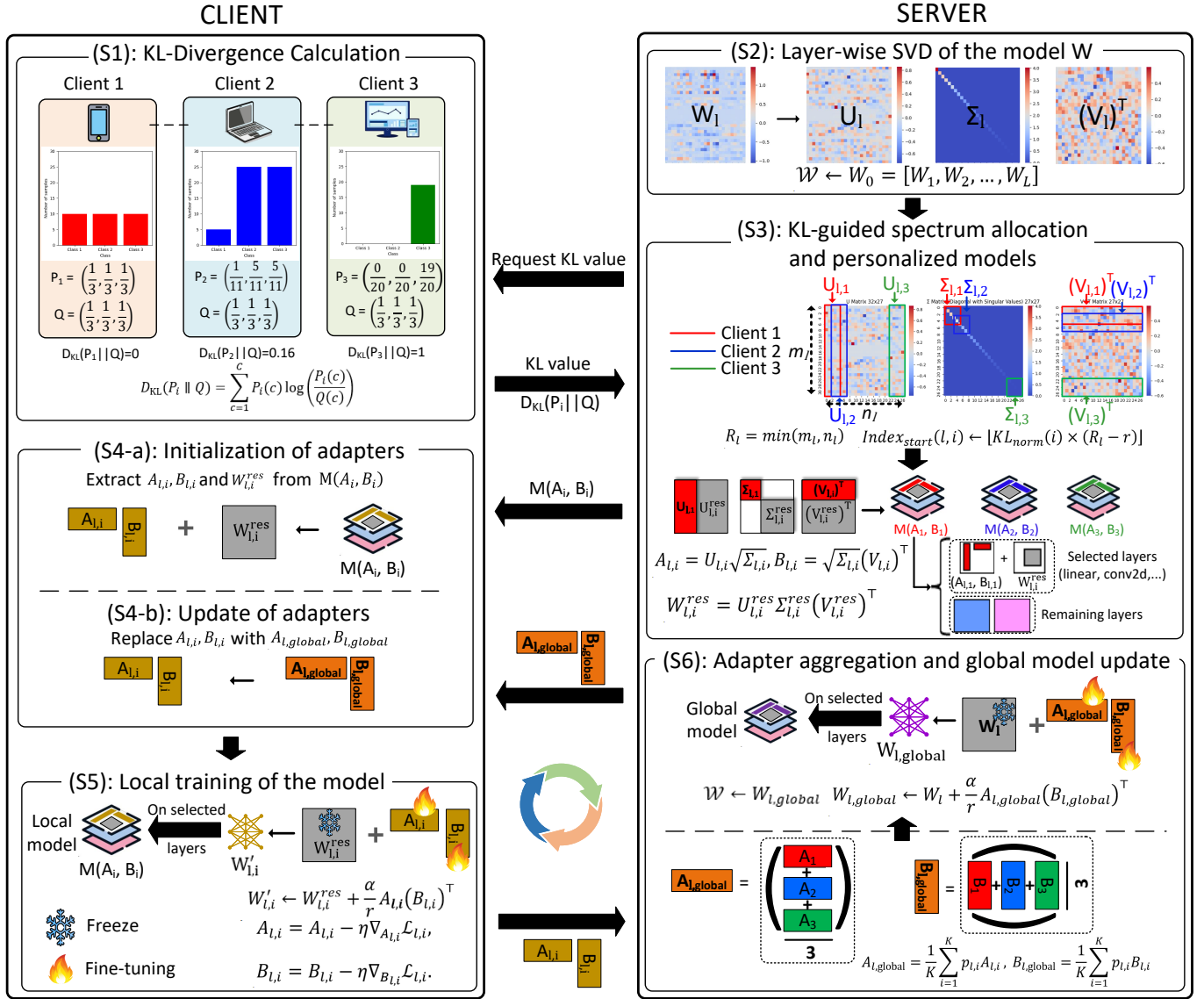


Fig. 2: Overall architecture of FedKLS.

initialization phase (or when data distribution needs to be refreshed), the server explicitly requests these KL values from clients, which are then transmitted back to the server. Based on these heterogeneity signals, the server performs a layer-wise SVD of the pretrained model and allocates different regions of the singular value spectrum to each client. Clients with near-IID data are assigned to principal spectral components, while highly non-IID clients are mapped to minor components, enabling structurally differentiated adaptation across clients.

Based on this allocation, the server constructs a personalized low-rank model for each client by instantiating adapters from the assigned spectral components. The model consists of selected layers equipped with low-rank adapters integrated with frozen residual weights, and remaining layers that are retained as part of the full model. This design enables coordinated adaptation in the shared spectral subspace while preserving local flexibility in the remaining layers, effectively balancing global knowledge sharing and client-specific learning.

Upon receiving this personalized model, each client initializes its local model with the provided low-rank components and frozen residual weights. During training, clients operate on this full model but update only the assigned low-rank adapters while keeping the residual weights fixed. The updated adapters are then aggregated at the server to form global adapters, which are redistributed to clients in subsequent rounds. This design enables heterogeneity-aware structural personalization by assigning clients to distinct spectral subspaces based on their KL divergence, thereby aligning global representation learning with client-specific adaptation.

### C. FedKLS Algorithm

We summarize the key notations used in FedKLS in Table I. Based on these definitions, Algorithm 1 describes the overall procedure of FedKLS, which consists of an Initialization/Reallocation Phase that captures data heterogeneity using

TABLE I: Table of notation for the FedKLS algorithm.

| Symbol                                       | Description                                                                               |
|----------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Data and Distribution</b>                 |                                                                                           |
| $C$                                          | Total number of classes from a dataset.                                                   |
| $n_i(c)$                                     | Number of samples of class $c$ in client $i$ .                                            |
| $N_i$                                        | Total number of samples in client $i$ .                                                   |
| $P_i(c)$                                     | Probability distribution of class $c$ in client $i$ 's local data.                        |
| $Q(c)$                                       | Reference (global or ideal IID) uniform distribution of class $c$ .                       |
| <b>KL Divergence and Indices</b>             |                                                                                           |
| $D_{\text{KL}}(P_i \parallel Q)$             | KL divergence between client $i$ 's distribution and ideal IID distribution.              |
| $KL_{\text{norm}}(i)$                        | Normalized KL divergence for client $i$ .                                                 |
| $D_{\text{KL}}^{\min}, D_{\text{KL}}^{\max}$ | Minimum and maximum KL divergence among clients.                                          |
| $\text{Index}_{\text{start}}(l, i)$          | Start index for chosen singular values in client $i$ 's low-rank adapter.                 |
| <b>Model and Weights</b>                     |                                                                                           |
| $L$                                          | Total number of selected layers in the model.                                             |
| $\mathcal{M}(A_i, B_i)$                      | Personalized low-rank models formed by adapters $A$ and $B$ at client $i$ .               |
| $W_{l,i}^{\text{res}}$                       | Frozen residual weight matrix at layer $l$ for client $i$ .                               |
| $W_{l,i}$                                    | Updated weight matrix at layer $l$ for client $i$ .                                       |
| $W_{l,\text{global}}$                        | Updated global weight matrix at layer $l$ after adapter aggregation.                      |
| $W_l$                                        | Base weight matrix at layer $l$ used for spectral decomposition and global model updates. |
| $W_0$                                        | Pretrained global model.                                                                  |
| <b>Adapters</b>                              |                                                                                           |
| $A_{l,i}, B_{l,i}$                           | Low-rank adapter matrices at layer $l$ for client $i$ .                                   |
| $A_{l,\text{global}}, B_{l,\text{global}}$   | Global adapter low-rank adapter matrices at layer $l$ .                                   |
| $R_l$                                        | Maximum rank of each selected layer $l$ .                                                 |
| $r$                                          | Selected rank of the low-rank adapters.                                                   |
| $\alpha$                                     | Scaling factor for low-rank adapters.                                                     |
| <b>Training</b>                              |                                                                                           |
| $\eta$                                       | Learning rate for adapter updates.                                                        |
| $\mathcal{L}_i$                              | Local loss function at client $i$ .                                                       |
| $\nabla_{A_i}, \nabla_{B_i}$                 | Gradients of the loss function with respect to adapters $A_i$ and $B_i$ .                 |
| $K$                                          | Total number of clients.                                                                  |
| $T$                                          | Total number of communication rounds.                                                     |
| $\tau$                                       | Optional update interval.                                                                 |

KL divergence and a Training Phase across the client and server sides.

At the client side, during the Initialization/Reallocation Phase (Lines 1—4), each client computes its local label distribution  $P_i(c)$  and the reference IID distribution  $Q(c)$ , followed by the KL divergence  $D_{\text{KL}}(P_i \parallel Q)$  to quantify statistical heterogeneity. Instead of transmitting the full distribution, each client sends only the scalar divergence value to the server, preserving data privacy while enabling cross-client comparison of statistical heterogeneity. This divergence reflects the degree of non-IIDness, where smaller values indicate distributions closer to IID and larger values correspond to more skewed or specialized data.

At the server side, during the Initialization/Reallocation Phase (Lines 12—24), the server sets the model weights  $\mathcal{W} \leftarrow W_0$  at initialization or updates  $\mathcal{W}$  using the latest global model  $W_{l,\text{global}}$  during reallocation. It then requests and receives KL divergence values from all clients and normalizes them to obtain  $KL_{\text{norm}}(i)$  with the range  $[0, 1]$ . Based on these normalized values, the server performs layer-wise singular value decomposition and assigns each client a contiguous spectral subspace by computing a client-specific starting index. From the selected components, low-rank adapters  $(A_{l,i}, B_{l,i})$  are constructed, while the remaining components form the frozen residual weights. The server assembles a personalized model  $\mathcal{M}(A_i, B_i)$  and sends it to each client.

During the client-side Training Phase (Lines 5—11), each client receives either the personalized model  $\mathcal{M}(A_i, B_i)$  (after initialization or reallocation) or the aggregated global adapters  $(A_{l,\text{global}}, B_{l,\text{global}})$  in subsequent rounds. The client initializes or updates its local adapters and residual weights accordingly. For each selected layer, the adapted weight is constructed as

$$W'_{l,i} \leftarrow W_{l,i}^{\text{res}} + \frac{\alpha}{r} A_{l,i} (B_{l,i})^\top, \quad (8)$$

where  $W_{l,i}^{\text{res}}$  remains fixed during local optimization. The client then performs local training by updating only the low-rank adapters  $(A_{l,i}, B_{l,i})$ , which reduces the number of trainable parameters and mitigates overfitting under heterogeneous data distributions. Since each client operates on a distinct spectral subspace determined during initialization, local updates remain specialized while preserving compatibility for aggregation. After training, the updated adapters  $(A_{l,i}, B_{l,i})$  are sent back to the server.

During the server-side Training Phase (Lines 25—28), the server receives updated adapters from clients and aggregates them to compute global adapters:

$$A_{l,\text{global}} = \frac{1}{K} \sum_{i=1}^K p_i A_{l,i}, \quad B_{l,\text{global}} = \frac{1}{K} \sum_{i=1}^K p_i B_{l,i}. \quad (9)$$

The global model is then updated as

$$W_{l,\text{global}} \leftarrow W_l + \frac{\alpha}{r} A_{l,\text{global}} (B_{l,\text{global}})^\top, \quad (10)$$

where  $W_l$  is kept fixed. The updated global adapters are redistributed to clients, completing the communication round.

The above procedure is repeated iteratively until convergence or a predefined number of communication rounds is reached. The reallocation phase, which collects KL divergence values at predefined intervals, can be optionally applied to accommodate time-varying data and dynamically reallocate low-rank subspaces. Although client models are reset during reallocation, they preserve accumulated knowledge from the latest global model, thereby maintaining robustness to distribution shifts.

#### D. Spectral Consistency of Cross-Slice Adapter Aggregation

##### a) Lemma 1 (Spectral Consistency of Aggregation):

Averaging adapter updates  $(A_{l,i}, B_{l,i})$  across clients results in a coordinate-wise aggregation with respect to the fixed spectral decomposition of  $W$ . Orthogonal spectral components do not interfere, and no cross-slice misalignment is introduced.

**Proof sketch.** Let  $W_l = U_l \Sigma_l V_l^\top$  denote the singular value decomposition of the pretrained weight matrix. Each client update  $(A_{l,i}, (B_{l,i})^\top)$  is restricted to a subset of spectral components defined by its assigned index set. Since all clients share the same singular vectors, these components form a common orthogonal basis. Therefore, aggregation combines coefficients only within aligned spectral directions, while non-overlapping components contribute zero. As a result, no cross-slice interference occurs.

**Implications.** Lemma 1 establishes that FedKLS can safely average adapter parameters  $(A_{l,i}, (B_{l,i})^\top)$  even when clients operate on different spectral slices. Although clients specialize in distinct regions of the singular value spectrum, all adaptive updates remain aligned under a common spectral decomposition, while unselected components are preserved in the frozen residual. As a result, averaging adapters from different subspaces is both theoretically correct and effective at retaining useful updates in the global model.



**Algorithm 1** FedKLS: KL-Guided Spectral Allocation for Federated Learning.

**Input:** pretrained model  $W_0$ , client set  $K$ , adapter rank  $r$ , optional update interval  $\tau$ .

**Output:** global adapters  $(A_{l,global}, B_{l,global})$  and personalized models  $\mathcal{M}(A_i, B_i)$  for each client  $i$ .

**Client side:**

**Initialization/Reallocation Phase**

- 1: Compute the actual local distribution  $P_i(c) = \frac{n_i(c)}{N_i}$ .
- 2: Compute the ideal IID distribution  $Q(c) = \frac{1}{C}$ .
- 3: Compute KL divergence  $D_{KL}(P_i \parallel Q) = \sum_{c=1}^C P_i(c) \log(\frac{P_i(c)}{Q(c)})$ .
- 4: Each client  $i$  sends their KL divergence  $D_{KL}(P_i \parallel Q)$  to the server.

**Training Phase**

- 5: Receive  $\mathcal{M}(A_i, B_i)$  at initialization/reallocation, or receive  $(A_{l,global}, B_{l,global})$  during training.
- 6: Initialize or update adapters:  $(A_{l,i}, B_{l,i}, W_{l,i}^{res}) \leftarrow \mathcal{M}(A_i, B_i)$  or  $(A_{l,i}, B_{l,i}) \leftarrow (A_{l,global}, B_{l,global})$ .
- 7: **for** each selected layer  $l$  **do**
- 8:   **Inject adapters:**  $W_{l,i}' \leftarrow W_{l,i}^{res} + \frac{\alpha}{r} A_{l,i} (B_{l,i})^\top$ , where  $W_{l,i}^{res}$  is frozen.
- 9:   **Local training:**  $A_{l,i} = A_{l,i} - \eta \nabla_{A_{l,i}} \mathcal{L}_{l,i}$ ,  $B_{l,i} = B_{l,i} - \eta \nabla_{B_{l,i}} \mathcal{L}_{l,i}$ .
- 10: **end for**
- 11: Send updated adapters  $(A_{l,i}, B_{l,i})$  back to the server.

**Server side:**

**Initialization/Reallocation Phase** (round  $t = 0$ , every  $\tau$  rounds if enabled)

- 12: At initialization,  $\mathcal{W} \leftarrow W_0 = [W_1, W_2, \dots, W_L]$ .
- 13: At reallocation,  $\mathcal{W} \leftarrow W_{l,global}$ .
- 14: Request and receive  $D_{KL}(P_i \parallel Q)$  from all clients.
- 15: **for** each client  $i$  **do**
- 16:   Normalize:  $KL_{norm}(i) \leftarrow \frac{D_{KL}(P_i \parallel Q) - D_{KL}^{min}}{D_{KL}^{max} - D_{KL}^{min}}$ .
- 17:   **for** each selected layer  $l$  in  $\mathcal{W}$  **do**
- 18:     Perform SVD on global weight:  $W_l = U_l \Sigma_l (V_l)^\top$ .
- 19:     Map to SVD index:  $Index_{start}(l, i) \leftarrow \lfloor KL_{norm}(i) \times (R_l - r) \rfloor$ .
- 20:     Construct adapters:  $A_{l,i} = U_{l,i} \sqrt{\Sigma_{l,i}}$ ,  $B_{l,i} = \sqrt{\Sigma_{l,i}} (V_{l,i})^\top$ .
- 21:     Compute residual matrices:  $W_{l,i}^{res} = U_{l,i} \Sigma_{l,i}^{res} (V_{l,i}^{res})^\top$ .
- 22:   **end for**
- 23:   Send  $\mathcal{M}(A_i, B_i)$ , including  $(A_{l,i}, B_{l,i})$ ,  $W_{l,i}^{res}$  and remaining layers weights, to client  $i$ .
- 24: **end for**

**Training Phase** (executed in all non-initialized/reallocation rounds)

- 25: Receive  $(A_{l,i}, B_{l,i})$  from all clients.
- 26: Server aggregates  $\{(A_{l,i}, B_{l,i})\}_{i=1}^K$  as  $A_{l,global} = \frac{1}{K} \sum_{i=1}^K p_{l,i} A_{l,i}$ ,  $B_{l,global} = \frac{1}{K} \sum_{i=1}^K p_{l,i} B_{l,i}$ .
- 27: Update global model:  $W_{l,global} \leftarrow W_l + \frac{\alpha}{r} A_{l,global} (B_{l,global})^\top$ , where  $W_l$  remains fixed during aggregation.
- 28: Server sends aggregated  $(A_{l,global}, B_{l,global})$  back to clients.

b) *Magnitude-driven weight contribution analysis:* Fig. 3 illustrates a clear disparity between principal and minor spectral components. As shown in Fig. 3c, the principal component matrix closely resembles the original weight matrix in Fig. 3a, indicating that most of the parameter energy is preserved in the dominant singular directions. In contrast, the minor component matrix in Fig. 3d exhibits substantially smaller values, concentrated in a narrow range (0-0.04), and shows limited structural similarity to the original weights.

**Key insight.** Since FedKLS constructs client adapters as linear combinations of these spectral components, updates derived from minor slices naturally have smaller magnitudes and therefore contribute less to the aggregated update at the server. Consequently, clients assigned to tail spectral regions—typically associated with higher distributional divergence—are implicitly down-weighted during aggregation, while high-energy, globally consistent directions dominate the

final model update. **Together, these observations confirm that averaging adapters across different spectral subspaces in FedKLS is both theoretically sound and practically effective, as the aggregation preserves important features, informative updates while suppressing low-impact, highly specialized perturbations.**

### E. Convergence Analysis

We analyze the convergence behavior of FedKLS and show that, despite KL-guided spectral personalization, it achieves the same order of convergence as standard federated optimization methods such as FedAvg, up to additional approximation terms introduced by low-rank and KL-guided subspace restriction.

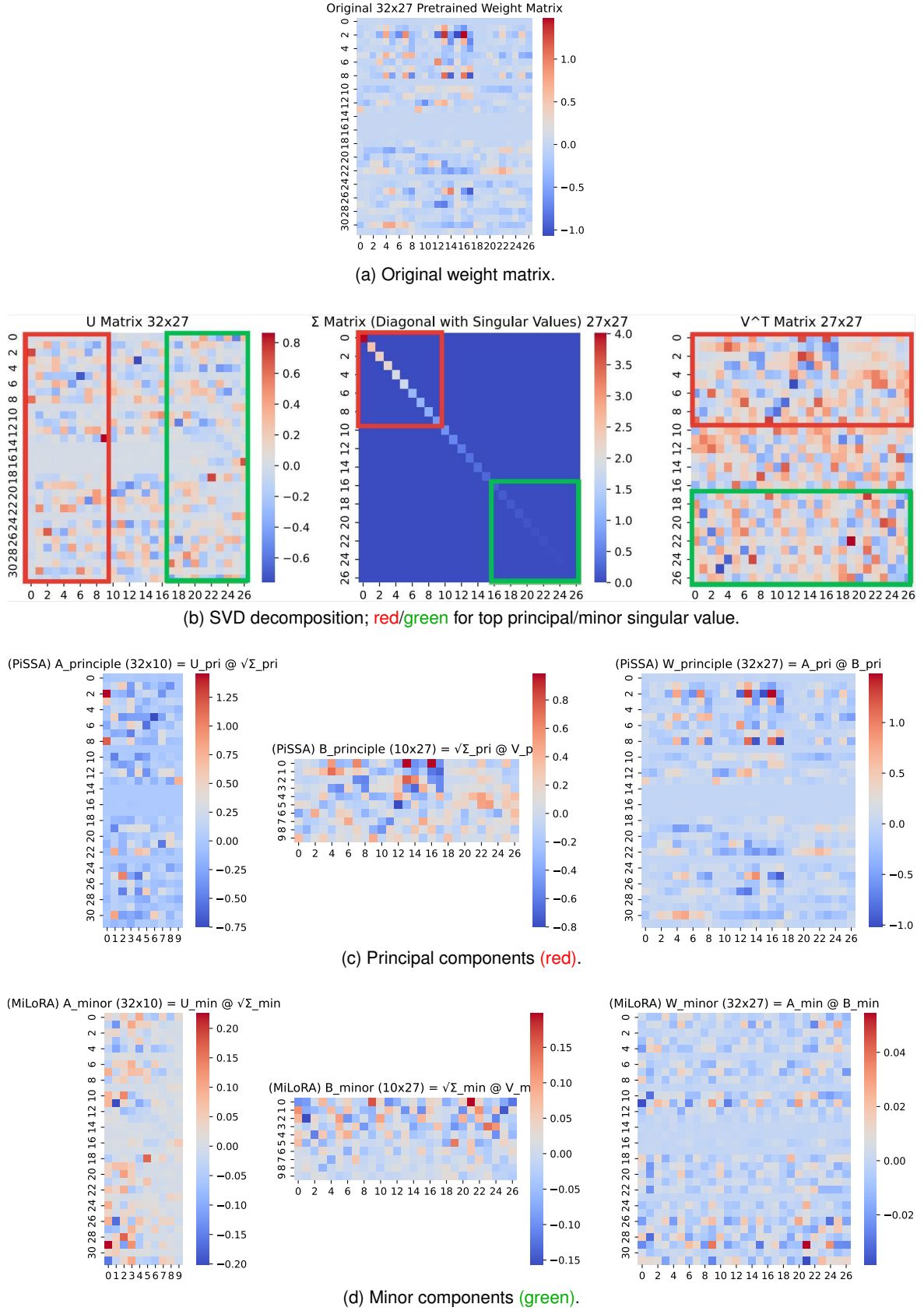


Fig. 3: Matrix visualizations of pretrained weights and their low-rank decompositions.

**Problem setup.** FedKLS optimizes a global objective

$$F(x) = \sum_{i=1}^K p_i F_i(x), \quad (11)$$

where  $K$  is the number of clients,  $p_i$  are normalized aggregation weights, and each  $F_i$  denotes the expected loss over client  $i$ 's data. The global model  $x$  is realized in practice via a shared pretrained backbone plus client-specific low-rank adapters  $\mathcal{M}_i = \mathcal{M}(A_i, B_i)$ , with  $A_i, B_i \in \mathbb{R}^{d \times r}$ , which are periodically averaged to update the global parameters.

**Assumptions.** We adopt standard assumptions used in non-convex federated learning:

- **Smoothness.** Each  $F_i$  is  $L$ -smooth:  $\|\nabla F_i(x) - \nabla F_i(y)\| \leq L\|x - y\|$  for all  $x, y$ .
- **Bounded variance.** Stochastic gradients satisfy  $\mathbb{E}\|\nabla f_i(x; \xi) - \nabla F_i(x)\|^2 \leq \sigma^2$ .
- **Bounded heterogeneity.**  $\|\nabla F_i(x) - \nabla F(x)\|^2 \leq \zeta^2$  for all  $i$  and  $x$ .
- **Low-rank approximation error.** For each selected layer  $l$ , let  $W_l^{(r)}$  denote the best rank- $r$  approximation of  $W_l$  under the Frobenius norm, and define

$$\Delta_{\text{SVD}} := \sum_l \|W_l - W_l^{(r)}\|_F^2. \quad (12)$$

This quantity upper bounds the squared bias in the gradients induced by restricting updates to the rank- $r$  subspace associated with  $W_l^{(r)}$ .

- **KL-guided subspace mismatch.** FedKLS assigns client-specific spectral slices based on normalized KL divergence, which may deviate from the optimal rank- $r$  subspace corresponding to  $W_l^{(r)}$ . Let  $W_{l, \mathcal{I}_i}$  denote the reconstruction of layer  $l$  using only the spectral components indexed by the KL-selected set  $\mathcal{I}_i$  for client  $i$ , i.e., the same components used to construct the adapter  $(A_{l,i}, B_{l,i})$ .

We define the averaged mismatch as

$$\Delta_{\text{KL}} := \sum_l \sum_{i=1}^K p_i \|W_l^{(r)} - W_{l, \mathcal{I}_i}\|_F^2, \quad (13)$$

which measures the deviation between the globally optimal rank- $r$  subspace and the client-specific spectral subspaces. This term captures the additional approximation bias introduced by KL-guided slice selection relative to the best rank- $r$  approximation.

**Convergence result.** Consider FedKLS run with step size  $\eta \leq 1/L$  for  $T$  communication rounds, and let  $x^{(t)}$  denote the (implicit) global parameter corresponding to the aggregated model at round  $t$ . Then the time-averaged squared gradient norm satisfies

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla F(x^{(t)})\|^2] &\leq \frac{2(F(x^{(0)}) - F^*)}{\eta T} + C_1 \eta L \sigma^2 \\ &\quad + C_2 \zeta^2 + C_3 \Delta_{\text{SVD}} + C_4 \Delta_{\text{KL}}, \end{aligned} \quad (14)$$

for some positive constants  $C_1, C_2, C_3, C_4$  that depend on the number of local steps and problem parameters, and where  $F^*$  is the minimum value of  $F$ .

**Interpretation.** The first term recovers the standard  $\mathcal{O}(1/T)$  decay in the averaged gradient norm as in FedAvg-style analyses for nonconvex objectives, while the second and third terms match the usual variance and heterogeneity contributions up to constants. The term  $\Delta_{\text{SVD}}$  reflects the intrinsic approximation error due to restricting optimization to a low-rank adapter space, and  $\Delta_{\text{KL}}$  captures the additional mismatch induced by KL-guided spectral slice selection. These terms appear additively, so they do not change the order of the convergence rate. FedKLS therefore does not claim to surpass the best rank- $r$  approximation; instead, it trades a controlled, explicitly quantified approximation bias for improved personalization across heterogeneous clients.

**Proof sketch.** The proof follows standard analyses of federated stochastic gradient methods.  $L$ -smoothness yields an expected descent inequality per round in terms of the true gradients and the stochastic gradient noise. The bounded-variance and bounded-heterogeneity assumptions control the contributions from stochastic sampling and client drift. Low-rank parameterization replaces the full gradients by their projections onto rank- $r$  subspaces, inducing a bounded gradient bias that can be upper bounded in terms of  $\Delta_{\text{SVD}}$ . KL-based subspace selection further perturbs these directions, causing an additional bias controlled by  $\Delta_{\text{KL}}$ . Combining these bounds and telescoping the descent inequality over  $T$  rounds yields the stated convergence result, where  $\Delta_{\text{SVD}}$  and  $\Delta_{\text{KL}}$  enter as additive approximation terms and do not affect the  $\mathcal{O}(1/T)$  convergence order.

## IV. EXPERIMENT

### A. Experimental Setup

We evaluate FedKLS on 20NewsGroup [23], Banking77 [24], and CIFAR10 [25], covering both text and vision modalities. Data are partitioned among clients using a Dirichlet distribution ( $\alpha \in \{0.05, 0.01\}$ ) [26] to simulate varying degrees of extreme statistical heterogeneity. Each client has access to its full local dataset from the beginning of training, and the data distribution remains fixed across all communication rounds. By default, KL-divergence values are computed once at initialization, unless periodic updates are explicitly specified.

For text classification, experiments are conducted with DistilBERT and Qwen1.5-0.5B on 20NewsGroup and Banking77, while CIFAR100 experiments are performed using ResNet18 for image classification. Adapters are applied to all linear layers in the Transformer architecture. For the vision model (Resnet), convolutional layers receive adapters only in the  $1 \times 1$  projection layers, following standard PEFT practice. All communicated adapter parameters are transmitted in FP32 unless otherwise noted. We compare FedKLS against parameter-efficient fine-tuning methods, including LoRA, PiSSA, MiLoRA, as well as full fine-tuning (FFT). As aggregation baselines, we adopt FedAvg, the standard averaging strategy in FL, and FedAWA [27], which adaptively adjusts aggregation weights using client update vectors to better handle data heterogeneity without relying on any proxy datasets.

TABLE II: F1 Score Performance on 20News and Banking77.

| Setting    |          |                      | $\alpha = 0.05$                    |                                    | $\alpha = 0.01$                    |                                    |
|------------|----------|----------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| Model      | Strategy | Method               | 20News                             | Banking77                          | 20News                             | Banking77                          |
| DistilBERT | FedAvg   | FFT                  | 0.8104 $\pm$ 0.003                 | 0.9476 $\pm$ 0.004                 | 0.8100 $\pm$ 0.006                 | 0.9489 $\pm$ 0.002                 |
|            |          | LoRA                 | 0.748 $\pm$ 0.013                  | 0.9328 $\pm$ 0.006                 | 0.7433 $\pm$ 0.007                 | 0.896 $\pm$ 0.001                  |
|            |          | PiSSA                | 0.7969 $\pm$ 0.004                 | 0.9442 $\pm$ 0.005                 | 0.8023 $\pm$ 0.001                 | 0.9455 $\pm$ 0.003                 |
|            |          | MiLoRA               | 0.7583 $\pm$ 0.014                 | 0.939 $\pm$ 0.005                  | 0.7579 $\pm$ 0.001                 | 0.9157 $\pm$ 0.007                 |
|            |          | <b>FedKLS (Ours)</b> | <b>0.8603<math>\pm</math>0.011</b> | <b>0.95<math>\pm</math>0.003</b>   | <b>0.8963<math>\pm</math>0.004</b> | <b>0.9638<math>\pm</math>0.009</b> |
|            | FedAWA   | FFT                  | 0.8135 $\pm$ 0.004                 | 0.9475 $\pm$ 0.002                 | 0.815 $\pm$ 0.003                  | 0.9461 $\pm$ 0.015                 |
|            |          | LoRA                 | 0.7475 $\pm$ 0.013                 | 0.9327 $\pm$ 0.008                 | 0.752 $\pm$ 0.005                  | 0.896 $\pm$ 0.005                  |
|            |          | PiSSA                | 0.7977 $\pm$ 0.004                 | 0.9448 $\pm$ 0.003                 | 0.804 $\pm$ 0.004                  | 0.941 $\pm$ 0.005                  |
|            |          | MiLoRA               | 0.763 $\pm$ 0.013                  | 0.9385 $\pm$ 0.006                 | 0.761 $\pm$ 0.007                  | 0.916 $\pm$ 0.007                  |
|            |          | <b>FedKLS (Ours)</b> | <b>0.8604<math>\pm</math>0.011</b> | <b>0.949<math>\pm</math>0.004</b>  | <b>0.894<math>\pm</math>0.003</b>  | <b>0.968<math>\pm</math>0.004</b>  |
| QWEN-0.5B  | FedAvg   | FFT                  | 0.8157 $\pm$ 0.005                 | 0.9238 $\pm$ 0.001                 | 0.822 $\pm$ 0.004                  | 0.917 $\pm$ 0.008                  |
|            |          | LoRA                 | 0.706 $\pm$ 0.009                  | 0.8623 $\pm$ 0.005                 | 0.711 $\pm$ 0.002                  | 0.854 $\pm$ 0.009                  |
|            |          | PiSSA                | 0.7664 $\pm$ 0.012                 | 0.8821 $\pm$ 0.003                 | 0.78 $\pm$ 0.005                   | 0.888 $\pm$ 0.006                  |
|            |          | MiLoRA               | 0.7362 $\pm$ 0.013                 | 0.8721 $\pm$ 0.013                 | 0.751 $\pm$ 0.016                  | 0.883 $\pm$ 0.0019                 |
|            |          | <b>FedKLS (Ours)</b> | <b>0.7705<math>\pm</math>0.005</b> | <b>0.8925<math>\pm</math>0.004</b> | <b>0.804<math>\pm</math>0.007</b>  | <b>0.896<math>\pm</math>0.009</b>  |
|            | FedAWA   | FFT                  | 0.7983 $\pm$ 0.001                 | 0.9259 $\pm$ 0.004                 | 0.825 $\pm$ 0.007                  | 0.938 $\pm$ 0.001                  |
|            |          | LoRA                 | 0.7015 $\pm$ 0.025                 | 0.8651 $\pm$ 0.07                  | 0.726 $\pm$ 0.008                  | 0.917 $\pm$ 0.018                  |
|            |          | PiSSA                | 0.7725 $\pm$ 0.016                 | 0.8983 $\pm$ 0.011                 | 0.781 $\pm$ 0.009                  | 0.92 $\pm$ 0.011                   |
|            |          | MiLoRA               | 0.7412 $\pm$ 0.021                 | 0.8818 $\pm$ 0.005                 | 0.762 $\pm$ 0.005                  | 0.905 $\pm$ 0.014                  |
|            |          | <b>FedKLS (Ours)</b> | <b>0.7825<math>\pm</math>0.011</b> | <b>0.9016<math>\pm</math>0.005</b> | <b>0.8115<math>\pm</math>0.011</b> | <b>0.935<math>\pm</math>0.025</b>  |

The main configuration consists of 10 clients with a 30% participation ratio, 1000 communication rounds, one local epoch per round, and a batch size of 32. We further conduct scalability experiments with 20, 50, and 100 clients. Learning rates are tuned from 0.01 to 0.0001 using SGD. The adapter rank is set to 32, scaling factor  $\alpha$  is equal to the rank. All experiments are implemented using the FedEasy framework [28] and executed on 4 $\times$ A100 GPUs for Qwen, CLIP ViT-B/32, and large-scale settings, and 4 $\times$ A6000 GPUS for DistilBERT and CIFAR10 experiments.

We report the maximum client-side F1-score and the minimum averaged loss across clients over 1000 training rounds, along with metrics including convergence behavior, total training time, the number of communication rounds required to reach the target performance, and the total communication cost. All results are averaged across three independent runs. To ensure a fair performance comparison among methods, the target F1-score for each configuration is defined based on the performance of LoRA under the most challenging heterogeneity setting. Specifically, for each dataset and model, we set the target threshold to the maximum F1-score achieved by LoRA, which consistently delivers the lowest performance among the PEFT baselines. In all tables, the best and second-best results among PEFT and personalized FL methods are highlighted in bold and underlined, respectively.

### B. Main results

1) *Language dataset*: Table II compares FedKLS against LoRA, PiSSA, MiLoRA, and FFT across two datasets: 20News and Banking77. FedKLS consistently delivers the best performance across both DistilBERT and Qwen models, with particularly large gains under highly non-IID settings ( $\alpha = 0.01$ ). For example, on 20News with DistilBERT under FedAvg, FedKLS achieves 0.8963 compared to 0.7433 for

LoRA and 0.8023 for PiSSA, while on Banking77 it reaches 0.9638 compared to 0.9455 (PiSSA). Even when compared with FFT, FedKLS remains competitive while requiring far fewer trainable parameters. These results confirm that KL-guided spectral selection strengthens low-rank adaptation by aligning updates with client-specific data distributions, leading to robust personalization and improved generalization.

Table III further evaluates communication efficiency in terms of convergence rounds, convergence time, and total communication cost until reach a target F1 score. The target F1 is set as the minimum score achieved by any method in each dataset-model pair, ensuring a fair comparison under consistent accuracy requirements. FedKLS significantly reduces communication overhead compared to FFT and other PEFT baselines while maintaining superior accuracy. For instance, on Banking77 with Qwen at  $\alpha = 0.01$ , FedKLS lowers the communication costs to 14.25 GB versus 196.098 GB for LoRA and 85.625 GB for MiLoRA. A similar trend is observed on Distilbert, where FedKLS converges in only 83.33 rounds compared to 795 rounds for LoRA. These results highlight that FedKLS not only improves accuracy but also scales efficiently in heterogeneous FL environments.

2) *Vision dataset*: Table IV reports the performance on CIFAR100 with ResNet18. FedKLS consistently achieves strong performance across both heterogeneity settings while maintaining high efficiency. Compared to PEFT baselines (LoRA, PiSSA, and MiLoRA), FedKLS attains the best F1 scores at both  $\alpha = 0.1$  (0.57) and  $\alpha = 0.01$  (0.589), while requiring substantially fewer communication rounds and lower training time. In particular, FedKLS converges in 305 and 265 rounds, significantly faster than LoRA (over 900 rounds) and MiLoRA (over 560 rounds), and also improves over PiSSA in both F1-score and efficiency.

Although FFT achieves the highest F1 score at  $\alpha = 0.1$ ,

TABLE III: Communication Efficiency on 20News and Banking77 Datasets.

| Setting   |                                                                                             |               | Round                     |                          | Time (hrs)               |                          | Cost (GB)       |                 |
|-----------|---------------------------------------------------------------------------------------------|---------------|---------------------------|--------------------------|--------------------------|--------------------------|-----------------|-----------------|
| Dataset   | Model                                                                                       | Method        | $\alpha = 0.05$           | $\alpha = 0.01$          | $\alpha = 0.05$          | $\alpha = 0.01$          | $\alpha = 0.05$ | $\alpha = 0.01$ |
| 20News    | DistilBERT<br>(Target F1: 0.73)<br>(Cost each round:<br>FFT: 1.0673 GB,<br>PEFT: 0.0648 GB) | FFT           | 88.33 $\pm$ 18.69         | 126.33 $\pm$ 6.23        | 0.321 $\pm$ 0.167        | 0.565 $\pm$ 0.017        | 141.97          | 203.05          |
|           |                                                                                             | LoRA          | 819.33 $\pm$ 124.68       | 795 $\pm$ 47.58          | 3.067 $\pm$ 0.946        | 3.026 $\pm$ 0.16         | 53.09           | 51.51           |
|           |                                                                                             | PiSSA         | 294.33 $\pm$ 14.05        | 326.33 $\pm$ 56.31       | 1.158 $\pm$ 0.097        | 1.242 $\pm$ 0.212        | 19.07           | 21.146          |
|           |                                                                                             | MiLoRA        | 563.66 $\pm$ 44.27        | 601.33 $\pm$ 38.42       | 2.167 $\pm$ 0.33         | 2.302 $\pm$ 0.151        | 36.52           | 38.966          |
|           |                                                                                             | <b>FedKLS</b> | <b>109.33</b> $\pm$ 29.94 | <b>83.33</b> $\pm$ 26.54 | <b>0.355</b> $\pm$ 0.2   | <b>0.246</b> $\pm$ 0.079 | <b>7.08</b>     | <b>5.4</b>      |
|           | QWEN-0.5B<br>(Target F1: 0.7)<br>(Cost each round:<br>FFT: 11.136 GB,<br>PEFT: 0.3654 GB)   | FFT           | 41 $\pm$ 12.14            | 41 $\pm$ 11.86           | 2.167 $\pm$ 0.136        | 2.159 $\pm$ 0.558        | 456.59          | 456.59          |
|           |                                                                                             | LoRA          | 796 $\pm$ 103.2           | 920.66 $\pm$ 59.77       | 35.67 $\pm$ 1.233        | 42.77 $\pm$ 2.547        | 290.85          | 336.41          |
|           |                                                                                             | PiSSA         | 105 $\pm$ 16.99           | 91.66 $\pm$ 7.039        | 4.258 $\pm$ 0.672        | 4.172 $\pm$ 0.383        | 38.36           | 33.495          |
|           |                                                                                             | MiLoRA        | 253 $\pm$ 39.25           | 293 $\pm$ 45.46          | 13.01 $\pm$ 0.788        | 13.295 $\pm$ 2.134       | 92.44           | 107.06          |
|           |                                                                                             | <b>FedKLS</b> | <b>93</b> $\pm$ 20.16     | <b>52.66</b> $\pm$ 5.312 | <b>2.502</b> $\pm$ 0.611 | <b>1.931</b> $\pm$ 0.21  | <b>33.98</b>    | <b>19.24</b>    |
| Banking77 | DistilBERT<br>(Target F1: 0.89)<br>(Cost each round:<br>FFT: 1.0673 GB,<br>PEFT: 0.0648 GB) | FFT           | 51 $\pm$ 4.966            | 68 $\pm$ 12.35           | 0.588 $\pm$ 0.021        | 0.752 $\pm$ 0.037        | 81.97           | 109.29          |
|           |                                                                                             | LoRA          | 568.66 $\pm$ 43.64        | 938.6 $\pm$ 23.29        | 3.892 $\pm$ 0.181        | 7.47 $\pm$ 0.877         | 36.84           | 60.82           |
|           |                                                                                             | PiSSA         | 270 $\pm$ 15.57           | 306.3 $\pm$ 7.717        | 2.251 $\pm$ 0.082        | 2.444 $\pm$ 0.661        | 12.96           | 19.85           |
|           |                                                                                             | MiLoRA        | 467 $\pm$ 39.73           | 704.3 $\pm$ 40.05        | 4.53 $\pm$ 0.17          | 5.267 $\pm$ 0.611        | 30.26           | 43.69           |
|           |                                                                                             | <b>FedKLS</b> | <b>199</b> $\pm$ 32.27    | <b>204</b> $\pm$ 27.17   | <b>0.298</b> $\pm$ 0.107 | <b>0.282</b> $\pm$ 0.133 | <b>15.12</b>    | <b>12.57</b>    |
|           | QWEN-0.5B<br>(Target F1: 0.8)<br>(Cost each round:<br>FFT: 11.136 GB,<br>PEFT: 0.3654 GB)   | FFT           | 20 $\pm$ 4.52             | 29.66 $\pm$ 5.312        | 0.675 $\pm$ 0.188        | 1.091 $\pm$ 0.176        | 222.72          | 330.37          |
|           |                                                                                             | LoRA          | 396 $\pm$ 56.89           | 536.66 $\pm$ 22.21       | 15.672 $\pm$ 0.866       | 17.015 $\pm$ 1.006       | 144.69          | 196.098         |
|           |                                                                                             | PiSSA         | 58 $\pm$ 22.65            | 78 $\pm$ 3.559           | 2.241 $\pm$ 0.551        | 2.4 $\pm$ 0.107          | 21.19           | 28.501          |
|           |                                                                                             | MiLoRA        | 195 $\pm$ 63.27           | 234.33 $\pm$ 20.99       | 6.72 $\pm$ 0.789         | 7.218 $\pm$ 0.672        | 71.25           | 85.625          |
|           |                                                                                             | <b>FedKLS</b> | <b>45</b> $\pm$ 25.63     | <b>39</b> $\pm$ 14.23    | <b>1.41</b> $\pm$ 0.151  | <b>1.238</b> $\pm$ 0.071 | <b>16.44</b>    | <b>14.25</b>    |

TABLE IV: Performance on CIFAR100 dataset.

| Setting  |                                                                                          |               | F1 score                |                          | Round                  |                       | Time (hrs)              |                         | Cost (GB)      |                 |
|----------|------------------------------------------------------------------------------------------|---------------|-------------------------|--------------------------|------------------------|-----------------------|-------------------------|-------------------------|----------------|-----------------|
| Dataset  | Model                                                                                    | Method        | $\alpha = 0.1$          | $\alpha = 0.01$          | $\alpha = 0.1$         | $\alpha = 0.01$       | $\alpha = 0.1$          | $\alpha = 0.01$         | $\alpha = 0.1$ | $\alpha = 0.01$ |
| CIFAR100 | ResNet18<br>(Target F1: 0.5)<br>(Cost each round:<br>FFT: 0.2687 GB,<br>PEFT: 0.0068 GB) | FFT           | 0.591 $\pm$ 0.053       | 0.583 $\pm$ 0.077        | 109 $\pm$ 8.11         | 173 $\pm$ 19.77       | 0.46 $\pm$ 0.03         | 0.488 $\pm$ 0.08        | 29.288         | 46.48           |
|          |                                                                                          | LoRA          | 0.513 $\pm$ 0.051       | 0.501 $\pm$ 0.047        | 912 $\pm$ 48.11        | 957 $\pm$ 73.11       | 2.867 $\pm$ 0.97        | 2.955 $\pm$ 0.88        | 6.201          | 6.507           |
|          |                                                                                          | PiSSA         | 0.567 $\pm$ 0.075       | 0.554 $\pm$ 0.134        | 318 $\pm$ 21.34        | 370 $\pm$ 25.78       | 1.132 $\pm$ 0.05        | 1.347 $\pm$ 0.09        | 2.162          | 2.516           |
|          |                                                                                          | MiLoRA        | 0.546 $\pm$ 0.073       | 0.542 $\pm$ 0.033        | 566 $\pm$ 54.32        | 582 $\pm$ 48.96       | 1.782 $\pm$ 0.12        | 1.999 $\pm$ 0.08        | 3.848          | 3.957           |
|          |                                                                                          | pFedEdit      | 0.565 $\pm$ 0.033       | 0.537 $\pm$ 0.022        | 425 $\pm$ 20.18        | 560 $\pm$ 18.67       | 1.218 $\pm$ 0.171       | 2.688 $\pm$ 0.221       | 10.85          | 12.57           |
|          | FedAS                                                                                    | FedAS         | 0.541 $\pm$ 0.077       | 0.535 $\pm$ 0.087        | 488 $\pm$ 16.33        | 623 $\pm$ 25.44       | 2.157 $\pm$ 0.11        | 2.518 $\pm$ 0.15        | 17.94          | 25.34           |
|          |                                                                                          | <b>FedKLS</b> | <b>0.57</b> $\pm$ 0.012 | <b>0.589</b> $\pm$ 0.016 | <b>305</b> $\pm$ 15.21 | <b>265</b> $\pm$ 5.11 | <b>1.032</b> $\pm$ 0.08 | <b>0.816</b> $\pm$ 0.05 | <b>2.074</b>   | <b>1.802</b>    |

it incurs significantly higher communication cost (29–46 GB) and does not offer advantages in convergence efficiency. In contrast, FedKLS reduces communication cost by more than an order of magnitude while maintaining competitive performance.

We further compare FedKLS with personalized and adaptive baselines, including pFedEdit [10] and FedAS [11]. While pFedEdit improves F1 over several PEFT methods, it requires substantially more communication rounds (425–560) and higher training time. FedAS shows moderate gains over MiLoRA but introduces additional overhead in both communication and computation. In comparison, FedKLS achieves better or comparable F1 scores while converging faster (305 and 265 rounds) and maintaining the lowest training time and communication cost among all compared methods.

Overall, these results demonstrate that FedKLS provides a favorable balance between accuracy, convergence speed, and system efficiency, particularly under heterogeneous data distributions.

3) *Multimodal dataset*: We evaluate the performance of FedKLS on the UMPC-Food101 dataset [29] under a highly non-IID federated setting, involving 20 clients with Dirichlet distribution ( $\alpha = 0.01$ ), learning rate 0.001, sample fraction 0.3, and FedAvg aggregation over 500 communication rounds,

where the target F1-score is set to 0.89. As the backbone model, we adopt the CLIP ViT-B/32 [18], which consists of a vision transformer image encoder pre-trained via contrastive language–image pretraining. All images are resized to 224 $\times$ 224 pixels.

As shown in Table V, FedKLS consistently outperforms all baselines in terms of accuracy, convergence, and communication efficiency. Specifically, FedKLS achieves the highest final F1-score of 0.9536, surpassing PiSSA and MiLoRA by a significant margin, while also maintaining the lowest loss. In addition, FedKLS converges substantially faster, requiring only 67 rounds to reach the target performance, compared to 81 rounds for PiSSA and 194 rounds for MiLoRA, and dramatically fewer rounds than LoRA. This accelerated convergence also translates into reduced training time, where FedKLS shortens the overall convergence time to 2.97 hours, nearly half of MiLoRA and significantly lower than other methods.

From a communication perspective, although most PEFT-based methods share similar per-round communication cost, FedKLS achieves the lowest total communication overhead due to faster convergence, reducing the total cost to 29.93 GB, which is approximately 8% lower than PiSSA and over 30% lower than MiLoRA. These results highlight that dynamically

TABLE V: Performance comparison on multimodal dataset UMPC-Food101.

| Method | F1-score (target = 0.89) |                     |                     | Loss              | Convergence       |                   | Time each round (s) | Cost            |                 |
|--------|--------------------------|---------------------|---------------------|-------------------|-------------------|-------------------|---------------------|-----------------|-----------------|
|        | Last                     | Max                 | Avg                 |                   | Round             | Time (h)          |                     | Cost/Round (MB) | Total Cost (GB) |
| FFT    | 0.9415 $\pm$ 0.001       | 0.9477 $\pm$ 0.001  | 0.9262 $\pm$ 0.0008 | 10.4 $\pm$ 0.57   | 22.6 $\pm$ 1.24   | 0.84 $\pm$ 0.07   | 150.8 $\pm$ 0.789   | 7441.29         | 168.66          |
| LoRA   | 0.8972 $\pm$ 0.008       | 0.9058 $\pm$ 0.004  | 0.6694 $\pm$ 0.03   | 56.419 $\pm$ 2.8  | 358.6 $\pm$ 6.64  | 14.219 $\pm$ 1.63 | 134.9 $\pm$ 9.508   | 442.33          | 158.64          |
| PiSSA  | 0.9207 $\pm$ 0.01        | 0.9337 $\pm$ 0.003  | 0.8756 $\pm$ 0.005  | 39.41 $\pm$ 0.722 | 81.33 $\pm$ 4.642 | 3.226 $\pm$ 0.398 | 141.26 $\pm$ 0.933  | 442.33          | 35.97           |
| MiLoRA | 0.9199 $\pm$ 0.002       | 0.9258 $\pm$ 0.0009 | 0.7775 $\pm$ 0.016  | 49.75 $\pm$ 2.116 | 194 $\pm$ 14.71   | 7.367 $\pm$ 0.46  | 141.38 $\pm$ 0.59   | 442.33          | 85.81           |
| FedKLS | 0.9499 $\pm$ 0.004       | 0.9512 $\pm$ 0.004  | 0.8751 $\pm$ 0.009  | 30.98 $\pm$ 2.01  | 67.66 $\pm$ 1.247 | 2.975 $\pm$ 0.063 | 140.61 $\pm$ 3.131  | 442.33          | 29.93           |

TABLE VI: Performance on 20News group and Banking77 dataset with streaming data.

| Dataset   | Setting                                                           |        | F1 score            |                    | Round               |                     | Time (hrs)         |                   | Cost (GB)       |                 |
|-----------|-------------------------------------------------------------------|--------|---------------------|--------------------|---------------------|---------------------|--------------------|-------------------|-----------------|-----------------|
|           | Model                                                             | Method | $\alpha = 0.05$     | $\alpha = 0.01$    | $\alpha = 0.05$     | $\alpha = 0.01$     | $\alpha = 0.05$    | $\alpha = 0.01$   | $\alpha = 0.05$ | $\alpha = 0.01$ |
| 20News    | Distilbert<br>(Target F1: 0.69)<br>(Interval $\tau$ : 10 rounds)  | FFT    | 0.795 $\pm$ 0.001   | 0.81 $\pm$ 0.007   | 84 $\pm$ 20.92      | 91 $\pm$ 7.874      | 0.697 $\pm$ 0.167  | 0.732 $\pm$ 0.056 | 135.01          | 146.26          |
|           |                                                                   | LoRA   | 0.7049 $\pm$ 0.002  | 0.7126 $\pm$ 0.01  | 860 $\pm$ 48        | 877 $\pm$ 27.58     | 6.55 $\pm$ 0.404   | 6.85 $\pm$ 0.209  | 55.72           | 56.82           |
|           |                                                                   | PiSSA  | 0.7639 $\pm$ 0.014  | 0.7823 $\pm$ 0.003 | 439 $\pm$ 33.13     | 357 $\pm$ 39.82     | 3.25 $\pm$ 0.265   | 2.67 $\pm$ 0.286  | 28.44           | 23.13           |
|           |                                                                   | MiLoRA | 0.7271 $\pm$ 0.02   | 0.7315 $\pm$ 0.014 | 752.33 $\pm$ 101.69 | 705.33 $\pm$ 69.139 | 5.703 $\pm$ 0.799  | 5.579 $\pm$ 0.428 | 48.751          | 45.705          |
|           |                                                                   | FedKLS | 0.8523 $\pm$ 0.005  | 0.8965 $\pm$ 0.007 | 219.33 $\pm$ 50.84  | 168.33 $\pm$ 15.32  | 1.08 $\pm$ 0.258   | 1.44 $\pm$ 0.516  | 14.21           | 10.908          |
| 20News    | Distilbert<br>(Target F1: 0.69)<br>(Interval $\tau$ : 100 rounds) | FFT    | 0.7992 $\pm$ 0.002  | 0.8081 $\pm$ 0.008 | 84 $\pm$ 25.31      | 74.66 $\pm$ 3.858   | 0.668 $\pm$ 0.195  | 0.647 $\pm$ 0.063 | 135.01          | 120.01          |
|           |                                                                   | LoRA   | 0.7195 $\pm$ 0.0007 | 0.6943 $\pm$ 0.004 | 821 $\pm$ 87        | 884 $\pm$ 26        | 6.297 $\pm$ 0.723  | 7.32 $\pm$ 0.299  | 53.2            | 57.28           |
|           |                                                                   | PiSSA  | 0.769 $\pm$ 0.009   | 0.7806 $\pm$ 0.007 | 423.3 $\pm$ 75.52   | 359 $\pm$ 25.46     | 2.963 $\pm$ 0.537  | 2.578 $\pm$ 0.191 | 27.432          | 23.263          |
|           |                                                                   | MiLoRA | 0.7281 $\pm$ 0.012  | 0.7285 $\pm$ 0.005 | 720.33 $\pm$ 50.6   | 739.66 $\pm$ 21.79  | 5.15 $\pm$ 0.371   | 5.99 $\pm$ 0.803  | 46.67           | 47.93           |
|           |                                                                   | FedKLS | 0.8475 $\pm$ 0.011  | 0.8866 $\pm$ 0.003 | 239.33 $\pm$ 36.55  | 171 $\pm$ 13.49     | 1.15 $\pm$ 0.134   | 0.82 $\pm$ 0.064  | 15.508          | 11.08           |
| Banking77 | Distilbert<br>(Target F1: 0.8)<br>(Interval $\tau$ : 10 rounds)   | FFT    | 0.942 $\pm$ 0.001   | 0.947 $\pm$ 0.003  | 61 $\pm$ 1.632      | 79.33 $\pm$ 2.494   | 0.334 $\pm$ 0.008  | 0.445 $\pm$ 0.009 | 98.04           | 127.51          |
|           |                                                                   | LoRA   | 0.8774 $\pm$ 0.006  | 0.8188 $\pm$ 0.004 | 803.66 $\pm$ 38.38  | 933 $\pm$ 36.12     | 3.789 $\pm$ 0.23   | 4.457 $\pm$ 0.267 | 52.077          | 60.458          |
|           |                                                                   | PiSSA  | 0.9362 $\pm$ 0.003  | 0.9228 $\pm$ 0.016 | 393.6 $\pm$ 37.54   | 364.6 $\pm$ 17.13   | 1.847 $\pm$ 0.083  | 1.726 $\pm$ 0.007 | 25.505          | 23.62           |
|           |                                                                   | MiLoRA | 0.9048 $\pm$ 0.001  | 0.8822 $\pm$ 0.006 | 627.33 $\pm$ 10.33  | 709.66 $\pm$ 16.21  | 2.89 $\pm$ 0.062   | 3.33 $\pm$ 0.061  | 40.65           | 45.98           |
|           |                                                                   | FedKLS | 0.9421 $\pm$ 0.001  | 0.9619 $\pm$ 0.011 | 303.66 $\pm$ 10.25  | 286.66 $\pm$ 12.88  | 0.914 $\pm$ 0.097  | 0.646 $\pm$ 0.09  | 19.677          | 18.576          |
| Banking77 | Distilbert<br>(Target F1: 0.8)<br>(Interval $\tau$ : 100 rounds)  | FFT    | 0.9461 $\pm$ 0.001  | 0.9455 $\pm$ 0.001 | 63.33 $\pm$ 2.494   | 76 $\pm$ 7.257      | 0.342 $\pm$ 0.017  | 0.419 $\pm$ 0.031 | 101.79          | 122.15          |
|           |                                                                   | LoRA   | 0.866 $\pm$ 0.007   | 0.823 $\pm$ 0.003  | 806 $\pm$ 55.87     | 949.33 $\pm$ 9.88   | 3.704 $\pm$ 0.278  | 4.734 $\pm$ 0.309 | 52.22           | 61.516          |
|           |                                                                   | PiSSA  | 0.9305 $\pm$ 0.002  | 0.9295 $\pm$ 0.008 | 383.33 $\pm$ 38.8   | 363.66 $\pm$ 17.98  | 1.89 $\pm$ 0.169   | 1.48 $\pm$ 0.051  | 24.83           | 23.56           |
|           |                                                                   | MiLoRA | 0.9114 $\pm$ 0.004  | 0.8767 $\pm$ 0.008 | 621 $\pm$ 37.17     | 753 $\pm$ 3.559     | 2.67 $\pm$ 0.149   | 3.61 $\pm$ 0.415  | 40.24           | 48.79           |
|           |                                                                   | FedKLS | 0.9385 $\pm$ 0.002  | 0.9641 $\pm$ 0.005 | 212.66 $\pm$ 18.2   | 208.33 $\pm$ 10.65  | 0.8818 $\pm$ 0.089 | 0.842 $\pm$ 0.324 | 13.78           | 13.49           |

allocating spectral subspaces based on client heterogeneity enables FedKLS to effectively balance global generalization and local personalization, leading to both improved performance and practical efficiency in federated IoT environments.

4) *Streaming dataset*: Table VI evaluates all methods under a data setting designed to evaluate time-varying data characteristics. At the beginning of training, each client is provided 30% of its local training data, and additional samples are gradually introduced at fixed intervals during training until the full dataset available. In this setting, periodic KL updates are enabled to reflect changes in client data distributions over time.

Across both 20News group and Banking77 datasets, FedKLS consistently achieves the highest F1 scores while requiring substantially fewer communication rounds, lower training time, and reduced communication cost compared to competing methods. Notably, the performance gap between FedKLS and other adapter-based approaches widens under smaller round steps (e.g., round step = 10), where the data distribution evolves more rapidly. This indicates that FedKLS adapts more effectively to changing data availability, benefiting from its KL-guided spectral allocation that prioritizes informative subspaces as new samples arrive. In contrast, LoRA, PiSSA, and MiLoRA exhibit slower convergence and higher system overhead, as their fixed or less adaptive parameterization struggles to accommodate the non-stationary training distribution. Overall, these results demonstrate that FedKLS is particularly well suited for streaming FL settings, maintaining strong accuracy while efficiently handling dynamic and incremental

data growth.

5) *Performance of FedKLS on each client*: Table VII and Fig 4 jointly provide a comprehensive view of the robustness of FedKLS under extreme non-IID settings ( $\alpha = 0.01$ ) with FedAvg strategy. The table highlights significant heterogeneity across clients in terms of sample size, number of classes, and KL divergence, while the figure illustrates the corresponding local training performance.

For the 20News dataset, clients with relatively IID distributions (i.e., low KL divergence), such as Client 3 (KL=0) and Client 0 (KL=0.3603), exhibit fast convergence and smooth optimization trajectories. These clients quickly reach stable and high F1-score levels, indicating that FedKLS preserves local consistency when data distributions are well aligned with the global objective. More importantly, FedKLS also performs effectively on highly non-IID clients. For instance, Client 4 shows extremely high KL divergence (KL\_norm=1), along with limited data and fewer classes. Although these clients initially suffer from unstable training and delayed convergence, their performance steadily improves and eventually approaches the level of more balanced clients. This demonstrates that the proposed global knowledge alignment mechanism effectively mitigates distribution mismatch and enables knowledge transfer from other clients.

A similar trend can be observed on the Banking77 dataset, which presents a larger label space and stronger heterogeneity. Clients with lower KL divergence, such as Client 5 and Client 9, achieve fast and stable convergence. In contrast, more heterogeneous clients, such as Client 1, exhibit slower progress

TABLE VII: Number of samples, classes, and KL\_norm for 20News and Banking77 clients under  $\alpha = 0.01$ .

| Client<br>(FedAvg) | 20News       |              |         | Banking77    |              |         |
|--------------------|--------------|--------------|---------|--------------|--------------|---------|
|                    | # of samples | # of classes | KL_norm | # of samples | # of classes | KL_norm |
| Client 0           | 1741         | 4            | 0.3603  | 1047         | 10           | 0.4209  |
| Client 1           | 128          | 2            | 0.9177  | 316          | 6            | 1       |
| Client 2           | 1652         | 3            | 0.4917  | 977          | 8            | 0.5582  |
| Client 3           | 2413         | 6            | 0       | 754          | 11           | 0.551   |
| Client 4           | 32           | 2            | 1       | 798          | 10           | 0.5452  |
| Client 5           | 1442         | 4            | 0.5238  | 1669         | 17           | 0       |
| Client 6           | 1068         | 5            | 0.8457  | 968          | 16           | 0.3717  |
| Client 7           | 1041         | 3            | 0.8714  | 1177         | 19           | 0.293   |
| Client 8           | 560          | 6            | 0.8337  | 872          | 23           | 0.4007  |
| Client 9           | 1237         | 7            | 0.6085  | 1425         | 28           | 0.1669  |

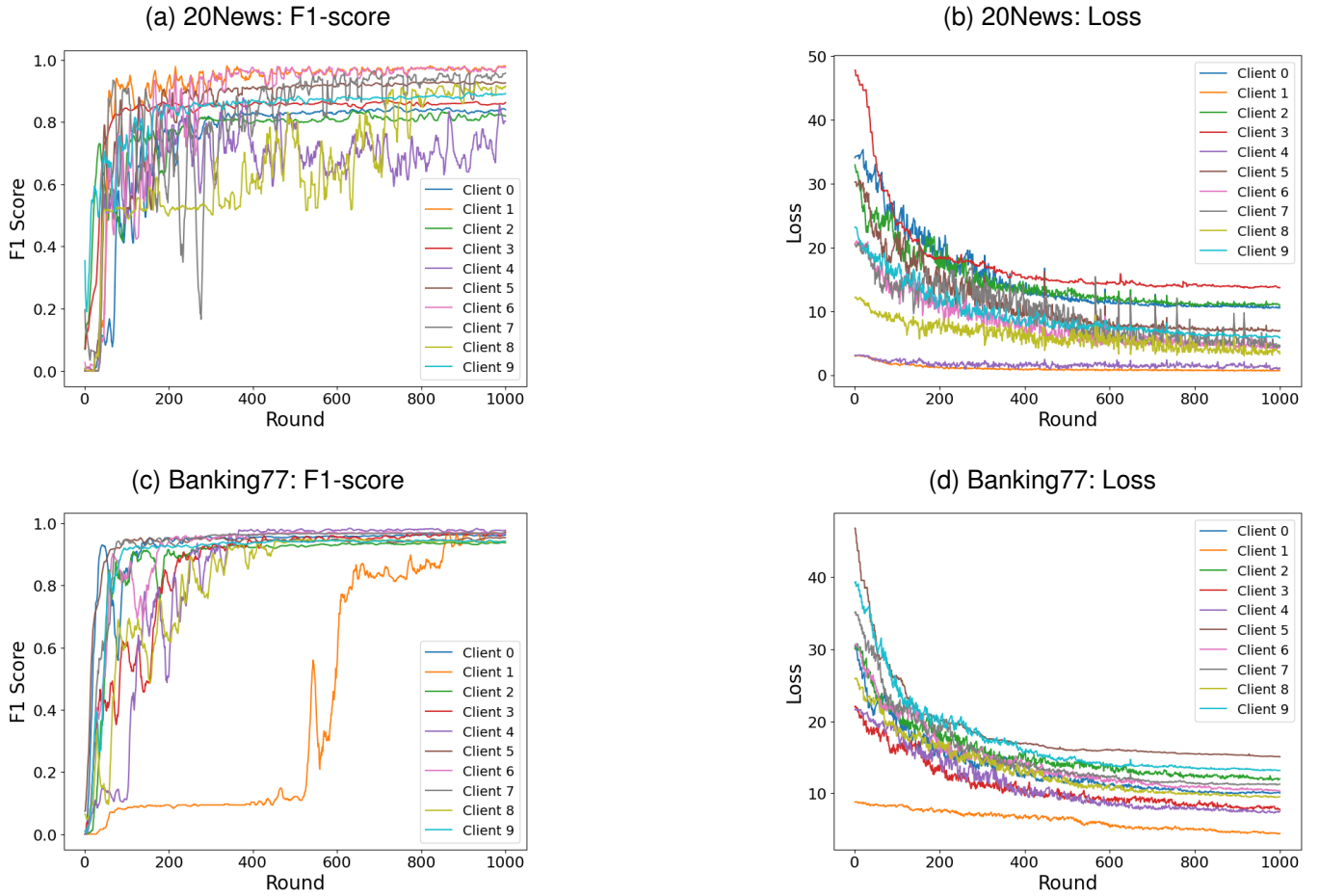


Fig. 4: Learning dynamics of FedKLS on individual clients under highly non-IID data ( $\alpha = 0.01$ ). Each curve corresponds to one client. The left column reports local F1-score and the right column shows local training loss across communication rounds.

in the early training stage but gradually improve as global information accumulates. This indicates that FedKLS reduces client drift and promotes consistent learning even in complex multi-class scenarios.

Furthermore, the loss curves show a monotonic and stable decrease across most clients for both datasets, without severe oscillations. Notably, both high-IID and high non-IID clients benefit from the proposed framework. For example, while low-divergence clients converge faster, highly heterogeneous clients also achieve competitive final performance, highlight-

ing the robustness and fairness of FedKLS across diverse data distributions.

Overall, these results confirm that FedKLS effectively balances local specialization and global knowledge sharing, enabling stable optimization and improved generalization across clients with varying levels of data heterogeneity.

6) *Results on Scaling the Number of Clients:* Table VIII shows how model performance scales as the number of clients increases from 20 to 100 with Dirichlet  $\alpha = 0.05$ . As expected, all methods degrade with more clients due to



TABLE VIII: Scaling clients.

| Setting  |               | 20 clients              |                         | 50 clients              |                         | 100 clients             |                         |
|----------|---------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Strategy | Method        | 20News                  | Banking77               | 20News                  | Banking77               | 20News                  | Banking77               |
| FedAvg   | FFT           | 0.820 $\pm$ 0.03        | 0.949 $\pm$ 0.08        | 0.811 $\pm$ 0.05        | 0.947 $\pm$ 0.02        | 0.804 $\pm$ 0.09        | 0.944 $\pm$ 0.07        |
|          | LoRA          | 0.701 $\pm$ 0.07        | 0.871 $\pm$ 0.06        | 0.694 $\pm$ 0.08        | 0.853 $\pm$ 0.07        | 0.595 $\pm$ 0.10        | 0.811 $\pm$ 0.09        |
|          | PiSSA         | 0.798 $\pm$ 0.04        | 0.883 $\pm$ 0.03        | 0.774 $\pm$ 0.05        | 0.865 $\pm$ 0.04        | <b>0.753</b> $\pm$ 0.06 | 0.831 $\pm$ 0.05        |
|          | MiLoRA        | 0.746 $\pm$ 0.06        | 0.867 $\pm$ 0.05        | 0.729 $\pm$ 0.07        | 0.851 $\pm$ 0.06        | 0.638 $\pm$ 0.09        | 0.823 $\pm$ 0.07        |
|          | FedKLS (Ours) | <b>0.866</b> $\pm$ 0.02 | <b>0.893</b> $\pm$ 0.02 | <b>0.800</b> $\pm$ 0.03 | <b>0.875</b> $\pm$ 0.03 | 0.695 $\pm$ 0.04        | <b>0.843</b> $\pm$ 0.03 |
| FedAWA   | FFT           | 0.819 $\pm$ 0.03        | 0.951 $\pm$ 0.07        | 0.809 $\pm$ 0.04        | 0.948 $\pm$ 0.03        | 0.805 $\pm$ 0.08        | 0.945 $\pm$ 0.06        |
|          | LoRA          | 0.703 $\pm$ 0.07        | 0.883 $\pm$ 0.06        | 0.700 $\pm$ 0.07        | 0.865 $\pm$ 0.06        | 0.595 $\pm$ 0.10        | 0.825 $\pm$ 0.08        |
|          | PiSSA         | 0.795 $\pm$ 0.04        | 0.895 $\pm$ 0.03        | 0.776 $\pm$ 0.05        | 0.877 $\pm$ 0.04        | <b>0.752</b> $\pm$ 0.06 | 0.838 $\pm$ 0.05        |
|          | MiLoRA        | 0.746 $\pm$ 0.06        | 0.879 $\pm$ 0.05        | 0.731 $\pm$ 0.07        | 0.861 $\pm$ 0.06        | 0.634 $\pm$ 0.09        | 0.835 $\pm$ 0.07        |
|          | FedKLS (Ours) | <b>0.867</b> $\pm$ 0.02 | <b>0.905</b> $\pm$ 0.02 | <b>0.800</b> $\pm$ 0.03 | <b>0.887</b> $\pm$ 0.03 | 0.694 $\pm$ 0.04        | <b>0.855</b> $\pm$ 0.03 |

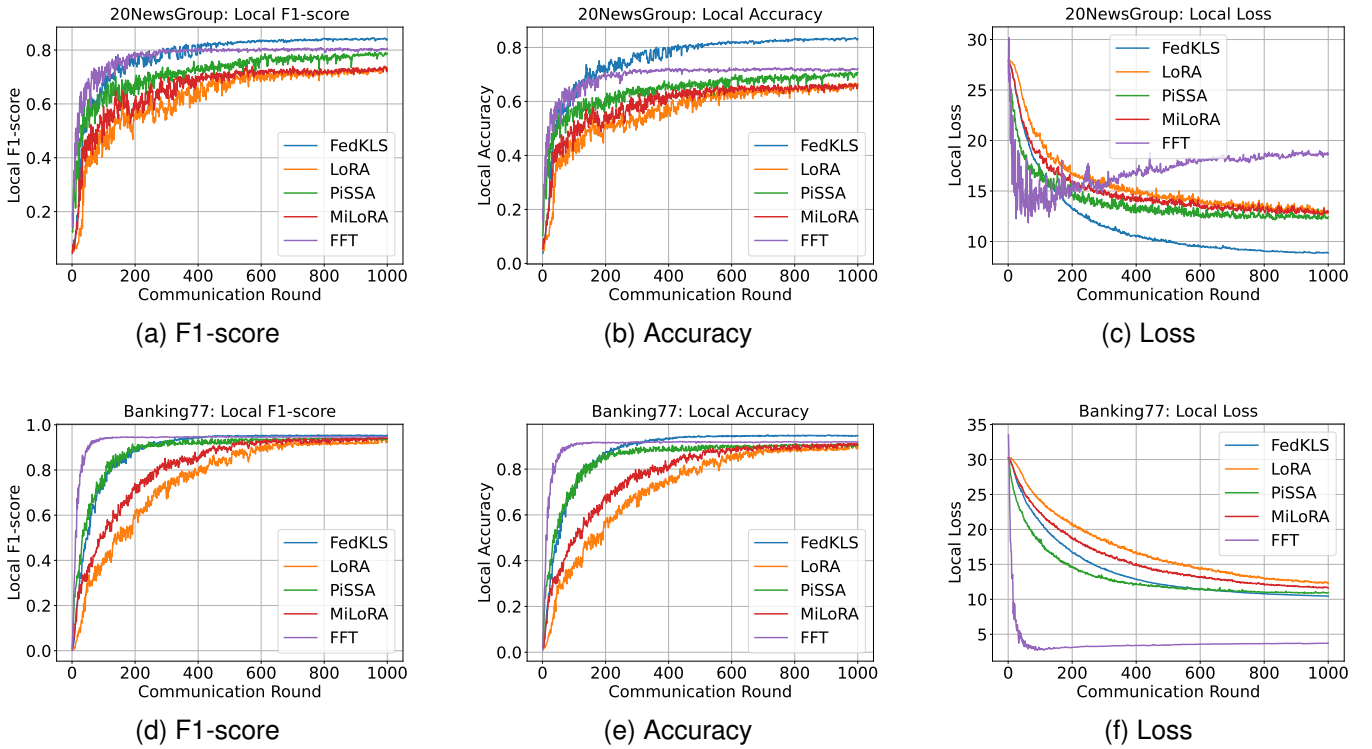


Fig. 5: Learning curves under non-IID data  $\alpha = 0.01$ . Top row: 20NewsGroup. Bottom row: Banking77. Columns show local F1-score, local accuracy, and local training loss across communication rounds.

rising heterogeneity, but FedKLS achieves the highest F1 score among PEFT baselines across most cases in both 20News and Banking77. PiSSA generally ranks second, while LoRA and MiLoRA drop more sharply under heavy skew. Compared to FedAvg and FedAWA with FFT, FedKLS offers the best trade-off by maintaining strong accuracy even in the 100-client case, highlighting the robustness of KL-guided subspace allocation for large-scale FL.

7) *Learning Curves of FedKLS under Extreme Non-IID Data:* Fig. 5 illustrates the learning curves of different adaptation methods on 20NewsGroup and Banking77 under extreme non-IID conditions. Across both datasets, FedKLS exhibits consistently faster convergence in local F1-score and accuracy, together with a more stable and monotonic reduction in local training loss. In contrast, baseline methods such as LoRA and

MiLoRA show slower improvement and pronounced oscillations, indicating sensitivity to heterogeneous client distributions.

On 20NewsGroup (top row), FedKLS rapidly improves local F1-score and accuracy within the early communication rounds while maintaining a smooth loss trajectory. This behavior suggests that KL-guided spectral allocation enables clients to focus their updates on more informative low-rank subspaces, thereby reducing conflicting gradients caused by severe label skew. PiSSA improves stability compared to LoRA but still converges more slowly and exhibits higher variance in the loss curve.

Similar trends are observed on Banking77 (bottom row), where FedKLS again achieves earlier stabilization and lower loss values. Notably, the gap between FedKLS and other low-



TABLE IX: Convergence and communication analysis of FedKLS under different ranks  $r$ .

| 20News (Target F1 = 0.78) |               |               |               |               |           |               |                    |                     |                 |
|---------------------------|---------------|---------------|---------------|---------------|-----------|---------------|--------------------|---------------------|-----------------|
| Rank $r$                  | Last          | Max           | Avg           | Loss          | ConvRound | ConvTime      | Time per Round (s) | Cost per Round (MB) | Total Cost (GB) |
| 1                         | 0.7128        | 0.7849        | 0.5786        | 17.778        | 907       | 5.9674        | <b>23.741</b>      | <b>3.1426</b>       | 2.8503          |
| 2                         | 0.8417        | 0.8645        | 0.6838        | 14.278        | 505       | 3.3325        | 23.858             | 5.181               | <b>2.616</b>    |
| 4                         | 0.8805        | 0.8886        | 0.7382        | 10.948        | 334       | 2.1909        | 23.906             | 9.2579              | 3.0921          |
| 8                         | 0.8895        | 0.8967        | 0.7905        | 9.2805        | 225       | 1.475         | 23.817             | 17.4116             | 3.9176          |
| 16                        | <u>0.9010</u> | 0.9029        | 0.8342        | 8.5737        | 103       | 0.6751        | 23.888             | 33.7191             | 3.473           |
| 32                        | <b>0.9015</b> | 0.9037        | 0.8589        | 8.268         | 72        | 0.4757        | 23.937             | 66.334              | 4.776           |
| 64                        | 0.8976        | 0.9006        | 0.8727        | 8.2791        | 50        | 0.3402        | 24.315             | 131.563             | 6.5781          |
| 128                       | 0.8962        | 0.9044        | 0.8816        | <b>8.179</b>  | <b>36</b> | <b>0.2534</b> | 24.883             | 262.023             | 9.4328          |
| 256                       | 0.8998        | <b>0.9052</b> | <b>0.8832</b> | <u>8.2333</u> | 42        | 0.3339        | 26.272             | 522.942             | 21.963          |

| Banking77 (Target F1 = 0.70) |               |               |               |               |           |               |                    |                     |                 |
|------------------------------|---------------|---------------|---------------|---------------|-----------|---------------|--------------------|---------------------|-----------------|
| Rank $r$                     | Last          | Max           | Avg           | Loss          | ConvRound | ConvTime      | Time per Round (s) | Cost per Round (MB) | Total Cost (GB) |
| 1                            | 0.6258        | 0.7061        | 0.5570        | 22.992        | 944       | 3.167         | <b>12.086</b>      | <b>3.1426</b>       | 2.9666          |
| 2                            | 0.7159        | 0.8018        | 0.6382        | 20.221        | 338       | 1.1385        | 12.2               | 5.181               | <b>1.7511</b>   |
| 4                            | 0.8421        | 0.8806        | 0.7136        | 17.243        | 233       | 0.7848        | 12.196             | 9.2579              | <u>2.157</u>    |
| 8                            | 0.9098        | 0.9197        | 0.7735        | 14.186        | 170       | 0.5755        | 12.178             | 17.4116             | 2.9599          |
| 16                           | 0.9396        | 0.9509        | 0.8397        | 12.073        | 105       | 0.3617        | 12.287             | 33.7191             | 3.5405          |
| 32                           | <b>0.9575</b> | 0.9596        | 0.8863        | 11.122        | 60        | 0.2094        | 12.333             | 66.334              | 3.98            |
| 64                           | 0.9558        | 0.9596        | 0.9084        | 10.576        | 40        | 0.143         | 12.42              | 131.563             | 5.2625          |
| 128                          | 0.9566        | <b>0.9599</b> | 0.9279        | 10.31         | 21        | <b>0.0815</b> | 12.916             | 262.023             | 5.5024          |
| 256                          | 0.9553        | <u>0.9582</u> | <b>0.9368</b> | <b>10.227</b> | <b>20</b> | <u>0.0856</u> | 13.919             | 522.9427            | 10.458          |

rank baselines becomes more pronounced in later rounds, indicating that FedKLS better preserves useful client-specific updates during aggregation rather than averaging away informative but heterogeneous signals. Overall, these learning curves confirm that FedKLS not only improves final performance but also enhances training stability and convergence efficiency under highly non-IID data distributions.

8) *Effect of Rank  $r$  in FedKLS*: Table IX analyzes the impact of the low-rank dimension  $r$  on the performance, convergence behavior, and communication cost of FedKLS on 20News and Banking77.

**F1-Score vs. Rank.** Increasing the rank  $r$  generally improves both the final and average F1-scores on both datasets. The most significant gains occur when moving from very low ranks (e.g.,  $r = 1, 2$ ) to moderate ranks (e.g.,  $r = 16, 32$ ), where the model gains sufficient representational capacity. Beyond this range, the improvements become marginal, indicating diminishing returns at higher ranks. This trend suggests that moderate ranks are sufficient to capture most of the useful task-specific information.

**Convergence Efficiency.** Larger ranks consistently accelerate convergence. As  $r$  increases, the number of communication rounds required to reach a target performance decreases substantially. This behavior indicates that higher-rank adapters provide stronger local update expressiveness, allowing faster alignment between local and global models.

**Communication-Computation Trade-off.** While increasing  $r$  reduces the number of communication rounds, it also increases the per-round communication cost due to larger adapter sizes. As a result, the total communication cost exhibits a non-monotonic (typically U-shaped) trend: it decreases initially due to faster convergence, but increases again when the per-round overhead dominates. The optimal trade-off is achieved at moderate ranks (e.g.,  $r \in 16, 32$ ), where both convergence speed and communication efficiency are balanced.

**Dataset Sensitivity.** Banking77 generally achieves higher

F1-scores and faster convergence than 20News at the same rank, likely due to its lower label complexity and more structured intent distribution. However, both datasets exhibit consistent rank-dependent trends, indicating that the observed behavior is robust across different tasks.

#### Key Takeaways.

- Increasing rank  $r$  improves accuracy and accelerates convergence, but with diminishing returns beyond moderate values.
- Very small ranks suffer from limited capacity, leading to slower convergence and lower performance.
- Very large ranks introduce significant communication overhead with limited additional benefit.
- A moderate rank (e.g.,  $r = 16$  or  $r = 32$ ) provides the best balance between F1-score, convergence speed, and communication efficiency.

## V. DISCUSSION AND LIMITATIONS

a) *Why KL Divergence for Client Heterogeneity Estimation*: FedKLS adopts KL divergence as a lightweight yet effective signal to quantify client heterogeneity and guide spectral subspace allocation. As summarized in Table X, alternative metrics such as Wasserstein distance and Fisher divergence can capture richer geometric or gradient-level discrepancies between client distributions; however, they typically require access to raw samples, intermediate embeddings, or model gradients, resulting in substantial computational overhead and increased privacy risks. In contrast, KL divergence can be computed efficiently using only local label histograms, incurs negligible computation cost, and preserves client privacy by avoiding exposure of raw data or learned representations. Moreover, FedKLS does not require an exact distributional match but only a monotonic heterogeneity signal to rank clients by non-IID severity and allocate appropriate spectral subspaces. Under this design, KL divergence provides a stable

TABLE X: Comparison of client heterogeneity metrics in federated learning.

| Metric            | What it measures                | Cost                               | Client req.           | Pros                     | Cons                     |
|-------------------|---------------------------------|------------------------------------|-----------------------|--------------------------|--------------------------|
| KL Divergence     | Label distribution mismatch     | <b>Very low</b> ( $O(\#classes)$ ) | Label counts only     | Simple, privacy-friendly | Ignores feature geometry |
| Wasserstein       | Geometric distribution mismatch | <b>High</b> ( $O(n^3)$ )           | Samples or embeddings | Captures geometry        | Expensive; privacy risk  |
| Fisher Divergence | Score-function mismatch         | <b>High</b> (density est.)         | Data + gradients      | Shape-sensitive          | Unstable in high-D       |

scalar measure that is sufficient for effective personalization while maintaining scalability. Exploring more expressive heterogeneity metrics that balance richness, efficiency, and privacy remains an interesting direction for future work.

*b) Privacy considerations of KL divergence sharing:*

FedKLS requires each client to transmit only a single scalar value—the KL divergence between its local label distribution and the reference distribution. This scalar contains substantially less information than typical federated updates such as gradients, logits, feature embeddings, or per-class statistics. Notably, the KL value does not reveal absolute class counts, exact class proportions, or any instance-level information. Moreover, the mapping from label histograms to KL divergence is many-to-one: distinct and semantically different label distributions can yield nearly identical KL values when evaluated against the same reference. As a result, the KL scalar conveys only a coarse notion of *how far* a client deviates from IID, but not *in which direction*. This loss of directional and class-specific information makes inversion mathematically underdetermined and significantly limits the feasibility of reconstruction or inference attacks, thereby providing a strong privacy–utility trade-off.

*c) Heuristic KL-to-Spectral Mapping:* FedKLS employs a fixed linear mapping from normalized KL divergence to contiguous spectral indices in order to allocate low-rank adaptation capacity across clients. This mapping is a deliberate design choice intended to provide a simple, stable, and interpretable baseline rather than a theoretically optimal assignment. While empirical diagnostics suggest that this heuristic preserves meaningful ordering between client heterogeneity and spectral allocation, it may not capture more complex or non-linear relationships that arise across models, layers, or data distributions. Designing adaptive or learned KL-to-spectrum mapping strategies that can adjust to such variations remains an important direction for future work.

*d) Choice of reference distribution:* In the current design, FedKLS adopts a uniform reference label distribution  $Q(c) = 1/C$  when computing client-side KL divergence. This choice is motivated by practicality and privacy: a uniform reference can be defined without collecting or estimating any global label statistics from clients, thereby preserving strict data isolation. We acknowledge that in real-world federated systems, the true global label distribution may be non-uniform, and a more faithful reference—such as a privately aggregated label histogram, secure aggregation protocol, or a small public proxy dataset—could better capture system-level imbalance. Importantly, the FedKLS framework itself is agnostic to the choice of  $Q(c)$  and is compatible with any reference distribution. Incorporating stronger and privacy-preserving estimators of the global distribution may further

improve spectral allocation accuracy, which we leave as an important direction for future work.

## VI. CONCLUSION

This paper introduced **FedKLS**, a FL framework that integrates KL-divergence-guided personalization with low-rank SVD-based adaptation. By mapping client-level distributional heterogeneity to distinct regions of the singular value spectrum, FedKLS constructs personalized LoRA-style adapters that balance global knowledge sharing with client-specific specialization. This spectral allocation mechanism enables efficient aggregation while mitigating the adverse effects of severe non-IID data.

Extensive experiments on 20NewsGroup, Banking77, CIFAR100, and UMPC-Food101 datasets using DistilBERT, Qwen, ResNet, and CLIP ViT-B/32 backbones demonstrate that FedKLS consistently outperforms state-of-the-art PEFT and personalized FL baselines in terms of accuracy, convergence speed, training stability, and communication efficiency. Notably, FedKLS maintains robust performance under extreme label skew and time-varying data streaming scenarios, highlighting its ability to preserve useful heterogeneous updates rather than averaging them away during aggregation.

Overall, FedKLS provides a principled and scalable approach to personalization in FL, offering strong empirical gains with minimal additional communication or privacy overhead. Future work will investigate adaptive spectral mapping strategies, secure estimation of global reference distributions, dynamic rank selection, and continually evolving federated environments.

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## REFERENCES

- [1] X. Li, K. Huang, W. Yang, S. Wang, and Z. Zhang, “On the convergence of fedavg on non-iid data,” *arXiv preprint arXiv:1907.02189*, 2019.
- [2] L. Collins, H. Hassani, A. Mokhtari, and S. Shakkottai, “Fedavg with fine tuning: Local updates lead to representation learning,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 10572–10586, 2022.
- [3] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen *et al.*, “Lora: Low-rank adaptation of large language models,” *International Conference on Learning Representations (ICLR)*, vol. 1, no. 2, p. 3, 2022.
- [4] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, “Qlora: Efficient finetuning of quantized llms,” *Advances in neural information processing systems*, vol. 36, pp. 10088–10115, 2023.

- [5] L. Zhang, L. Zhang, S. Shi, X. Chu, and B. Li, "Lora-fa: Memory-efficient low-rank adaptation for large language models fine-tuning," *arXiv preprint arXiv:2308.03303*, 2023.
- [6] X. Ma, J. Zhu, Z. Lin, S. Chen, and Y. Qin, "A state-of-the-art survey on solving non-iid data in federated learning," *Future Generation Computer Systems*, vol. 135, pp. 244–258, 2022.
- [7] X. Yuan and P. Li, "On convergence of fedprox: Local dissimilarity invariant bounds, non-smoothness and beyond," *Advances in Neural Information Processing Systems*, vol. 35, pp. 10752–10765, 2022.
- [8] S. P. Karimireddy, S. Kale, M. Mohri, S. Stich, and A. T. Suresh, "Scaffold: Stochastic controlled averaging for federated learning," *International conference on machine learning*, pp. 5132–5143, 2020.
- [9] Q. Li, B. He, and D. Song, "Model-contrastive federated learning," *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 10713–10722, 2021.
- [10] H. Yuan, W. Paul, J. Aucott, P. Burlina, and Y. Cao, "Pfededit: Personalized federated learning via automated model editing," *European Conference on Computer Vision*, pp. 91–107, 2024.
- [11] X. Yang, W. Huang, and M. Ye, "Fedas: Bridging inconsistency in personalized federated learning," *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 11986–11995, 2024.
- [12] C. T. Dinh, N. Tran, and J. Nguyen, "Personalized federated learning with moreau envelopes," *Advances in neural information processing systems*, vol. 33, pp. 21394–21405, 2020.
- [13] M. G. Arivazhagan, V. Aggarwal, A. K. Singh, and S. Choudhary, "Federated learning with personalization layers," *arXiv preprint arXiv:1912.00818*, 2019.
- [14] T. Li, S. Hu, A. Beirami, and V. Smith, "Ditto: Fair and robust federated learning through personalization," *International conference on machine learning*, pp. 6357–6368, 2021.
- [15] S. Targ, D. Almeida, and K. Lyman, "Resnet in resnet: Generalizing residual architectures," *arXiv preprint arXiv:1603.08029*, 2016.
- [16] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter," *arXiv preprint arXiv:1910.01108*, 2019.
- [17] J. Bai, S. Bai, Y. Chu, Z. Cui, K. Dang, X. Deng, Y. Fan, W. Ge, Y. Han, F. Huang *et al.*, "Qwen technical report," *arXiv preprint arXiv:2309.16609*, 2023.
- [18] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," *International Conference on Learning Representations (ICLR)*, 2021.
- [19] S.-Y. Liu, C.-Y. Wang, H. Yin, P. Molchanov, Y.-C. F. Wang, K.-T. Cheng, and M.-H. Chen, "Dora: Weight-decomposed low-rank adaptation," *Forty-first International Conference on Machine Learning*, 2024.
- [20] D. J. Kopiczko, T. Blankevoort, and Y. M. Asano, "Vera: Vector-based random matrix adaptation," *arXiv preprint arXiv:2310.11454*, 2023.
- [21] Q. Zhang, M. Chen, A. Bukharin, N. Karampatziakis, P. He, Y. Cheng, W. Chen, and T. Zhao, "Adalora: Adaptive budget allocation for parameter-efficient fine-tuning," *arXiv preprint arXiv:2303.10512*, 2023.
- [22] S. Yun, S. Chae, D. Lee, and Y. Ro, "Soma: Singular value decomposed minor components adaptation for domain generalizable representation learning," *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 25602–25612, 2025.
- [23] K. Borkar and N. Dhande, "Efficient text classification of 20 newsgroup dataset using classification algorithm," *Int J Recent Innov Trends Comput Commun*, vol. 5, no. 6, pp. 1236–1240, 2017.
- [24] V. Srivastava, "Lending an ear: How llms hear your banking intentions," *Proceedings of the 5th ACM International Conference on AI in Finance*, pp. 301–309, 2024.
- [25] Y. Abouelnaga, O. S. Ali, H. Rady, and M. Moustafa, "Cifar-10: Knn-based ensemble of classifiers," *2016 International Conference on Computational Science and Computational Intelligence (CSCI)*, pp. 1192–1195, 2016.
- [26] Y. Wang, Y. Tong, and D. Shi, "Federated latent dirichlet allocation: A local differential privacy based framework," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, no. 04, pp. 6283–6290, 2020.
- [27] C. Shi, H. Zhao, B. Zhang, M. Zhou, D. Guo, and Y. Chang, "Fedawa: Adaptive optimization of aggregation weights in federated learning using client vectors," *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 30651–30660, 2025.
- [28] M. Kundroo, G. Haider, N. Khoa, A. W. Mamond, and T. Kim, "Fedeasy: Federated learning with ease," *SoftwareX*, vol. 31, p. 102276, 2025.

- [29] L. Bossard, M. Guillaumin, and L. Van Gool, "Food-101—mining discriminative components with random forests," *European conference on computer vision*, pp. 446–461, 2014.

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